VajraKavach

Secure Job Search & Professional Networking Platform

User Guide

Comprehensive Platform Usage Manual

CSE 345/545: Foundations to Computer Security

Version	1.0
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Platform URL	`https://192.168.3.44`
Audience	Users, Recruiters, Admins
Course	CSE 345/545 — IIIT-Delhi

This guide covers all platform features for the three user roles:
User (Job Seeker), Recruiter, and Admin.

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Introduction

VajraKavach is a **Secure Job Search & Professional Networking Platform** built for the CSE 345/545 course at IIIT-Delhi. It provides a LinkedIn-like experience with enterprise-grade security features including:

- **TOTP-based Two-Factor Authentication** with a virtual keyboard to prevent keyloggers
- **End-to-End Encrypted (E2EE) Messaging** using RSA-4096 + AES-256-GCM hybrid encryption
- **Encrypted Resume Storage** with TOTP-gated downloads
- **PKI-backed Trust** for message signing and resume integrity
- **Tamper-Evident Audit Logs** with hash chaining and optional blockchain
- **Role-Based Access Control (RBAC)** across four roles

User Roles

The platform supports three user-facing roles:

Role	Capabilities
User (Job Seeker)	Create and edit profile, upload/download resumes, search and apply for jobs, track applications, send/receive encrypted messages, manage connections, view connection graph
Recruiter	All User capabilities plus: create company pages, post and manage job listings, view and manage applicants, update application statuses, message candidates
Admin	All User capabilities plus: access admin dashboard (**view-only mode**), view all users, view all job listings, view resumes metadata, view audit logs, verify hash chain integrity, view blockchain blocks, view user reports. *Note: Admin cannot suspend/delete users, delist jobs, create announcements, mine blockchain blocks, or resolve reports — those actions require super admin (Platform Admin) privileges which are not available through registration.*

Your role is assigned during registration. Roles cannot be changed after registration. If you need to change the roles, delete your account and create new account.
The **Admin** role available during registration gives you **read-only access** to the admin dashboard. Full administrative privileges (suspending users, deleting accounts, delisting jobs, creating announcements, mining blockchain, resolving reports) are reserved for the **Platform Admin** (super admin) role, which is not available through self-registration.

Getting Started

System Requirements

- A modern web browser (Chrome, Firefox, Edge, Safari)
- Network access to `192.168.3.0/24` subnet (LAN connectivity)
- A TOTP authenticator app installed on your phone:
  - **Recommended:** Google Authenticator, Microsoft Authenticator, Authy, or FreeOTP
- The VajraKavach root CA certificate installed on your device (see Section 2.2)

Certificate Installation

VajraKavach uses a self-signed TLS certificate for HTTPS. To avoid browser security warnings, you must install the root CA certificate (`rootCA.pem`) on your device.

△ Important

Without the certificate installed, your browser will display a "Your connection is not private" warning every time you visit the platform. Installing the certificate removes this warning and enables full HTTPS trust.

Obtaining the Certificate

The certificate file `rootCA.pem` is distributed via Google Drive. Link to certificate file : Link to certificate file

1. Open the shared Google Drive link provided to you
2. Download the file named `rootCA.pem`
3. Save it to a known location on your device (e.g., Downloads folder)

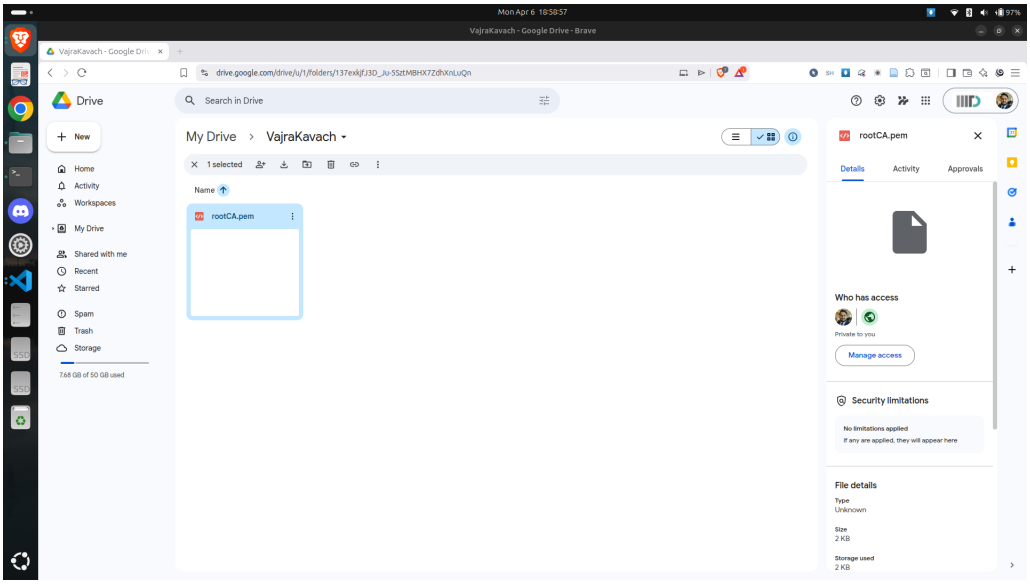


Figure 1: Google Drive link showing the rootCA.pem file available for download

Installing on Linux (Ubuntu/Debian)

1. Open a terminal
2. Copy the certificate to the trusted store:

```
sudo cp rootCA.pem /usr/local/share/ca-certificates/
   vajrakavach-rootCA.crt
```

3. Update the certificate store:

```
sudo update-ca-certificates
```

4. **For browser trust (Chrome/Chromium):**
   (a) Open Chrome and navigate to `chrome://settings/certificates`
   (b) Click the **Authorities** tab
   (c) Click **Import**, select the `rootCA.pem` file
   (d) Check **"Trust this certificate for identifying websites"**
   (e) Click **OK**
5. **For Firefox:**
   (a) Open Firefox and navigate to `about:preferences#privacy`
   (b) Scroll to **Certificates** and click **View Certificates**
   (c) Under **Authorities**, click **Import**
   (d) Select the `rootCA.pem` file
   (e) Check **"Trust this CA to identify websites"** and click **OK**

Installing on Windows

1. Double-click the downloaded `rootCA.pem` file
2. In the Certificate dialog, click **Install Certificate...**
3. Select **Local Machine** (requires Administrator) or **Current User**
4. Click **Next**
5. Select **"Place all certificates in the following store"**
6. Click **Browse**, select **"Trusted Root Certification Authorities"**
7. Click **Next**, then **Finish**
8. Accept the security warning prompt
9. Restart your browser

On Windows, Chrome and Edge automatically use the system certificate store. Firefox uses its own store — if you use Firefox on Windows, follow the Firefox steps from the Linux section above.

Installing on macOS

1. Double-click the downloaded `rootCA.pem` file to open it in **Keychain Access**
2. Select the **System** keychain (or **login** keychain)
3. Click **Add**
4. Find the certificate in the keychain, double-click it
5. Expand the **Trust** section
6. Set **"When using this certificate"** to **"Always Trust"**
7. Close the dialog and enter your macOS password to confirm
8. Restart your browser

Installing on Android

1. Download `rootCA.pem` from the Google Drive link on your Android device
2. Go to **Settings** → **Security** (or **Security & Privacy**)
3. Tap **Encryption & credentials** (or **More security settings**)
4. Tap **Install a certificate** → **CA certificate**
5. Tap **Install anyway** when warned

6. Browse to and select the downloaded `rootCA.pem` file
7. The certificate will be installed and visible under **Trusted credentials** → **User** tab

> △ Important
>
> The exact path varies by Android version and manufacturer. On Samsung devices, look under **Settings** → **Biometrics and Security** → **Other security settings** → **Install from device storage**. On Pixel devices, look under **Settings** → **Security** → **More security settings** → **Encryption & credentials**.

Installing on iOS / iPadOS

1. Download `rootCA.pem` from the Google Drive link on your iOS device
2. A prompt will appear saying **"Profile Downloaded"** — tap **Close**
3. Go to **Settings** → **General** → **VPN & Device Management**
4. Under **Downloaded Profile**, tap the VajraKavach certificate
5. Tap **Install** (enter your device passcode)
6. Tap **Install** again on the warning screen
7. Go to **Settings** → **General** → **About** → **Certificate Trust Settings**
8. Enable full trust for the VajraKavach root certificate

Accessing the Platform

Once the certificate is installed:

1. Open your browser
2. Navigate to `https://192.168.3.44`
3. The landing page should load without any certificate warnings
4. You should see the VajraKavach landing page with options to **Login/Sign in** or **Register**

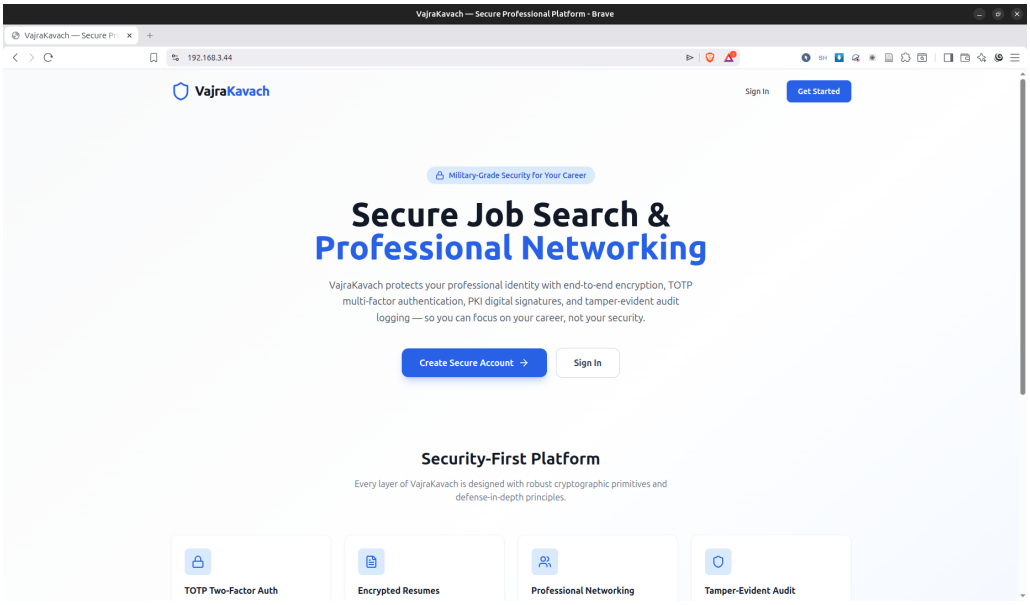


Figure 2: VajraKavach landing page showing the platform name, description, and Login/Register buttons

Registration & Account Setup

Creating an Account

1. From the landing page, click the **Get Started** button (or navigate to `/register`)
2. Fill in the registration form:
   - **Full Name** — your display name on the platform
   - **Email Address** — must be unique; used for login
   - **Password** — must meet the following requirements:
     - 8–72 characters
     - At least one uppercase letter
     - At least one lowercase letter
     - At least one digit
     - At least one special character

     A **password strength indicator** shows which requirements are met.
   - **Phone** (optional) — select a country code (e.g., IN +91, US +1) and enter your phone number
   - **Role** — select one of: **Job Seeker**, **Recruiter**, or **Admin (Read-only)**
3. Click **Register**

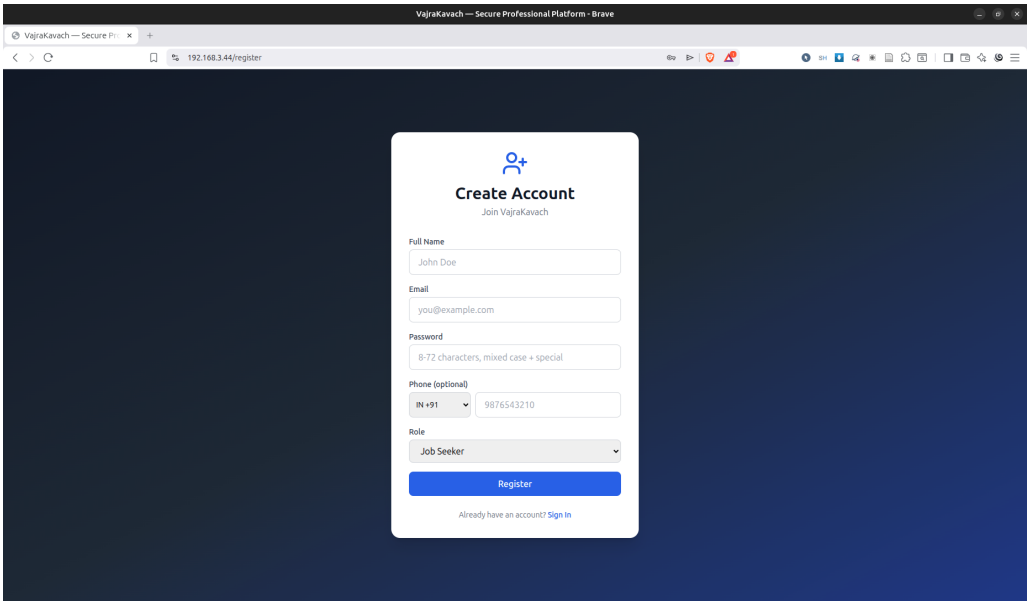


Figure 3: Registration form with fields for name, email, password, confirm password, and role selection

Setting Up TOTP (Two-Factor Authentication)

After clicking Register, you will be presented with a TOTP setup screen. This is **mandatory** — you cannot use the platform without TOTP.

1. A **QR code** will appear on screen
2. Open your authenticator app (e.g., Google Authenticator)
3. Tap the + button in your authenticator app and select **Scan QR Code**
4. Scan the QR code displayed on the VajraKavach screen
5. Your authenticator app will now generate a 6-digit code that changes every 30 seconds
6. Enter the current 6-digit code using the **virtual keyboard** displayed on screen
7. Click **Verify** to complete the setup

Save your TOTP secret! If you lose access to your authenticator app, you will need the TOTP secret to recover your account. Some authenticator apps (like Authy) support cloud backup. Google Authenticator can export codes via QR transfer to a new device.

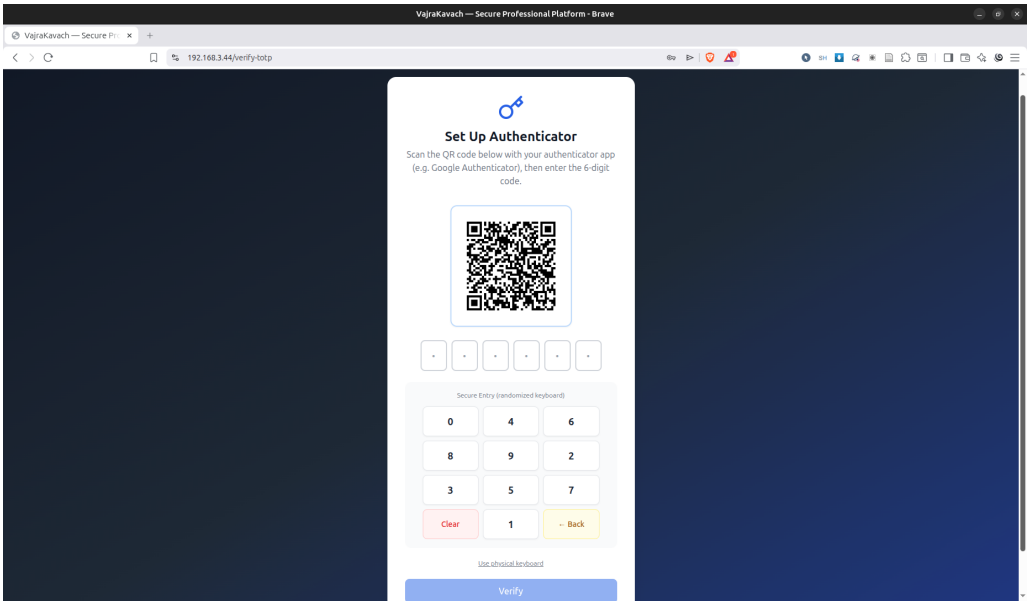


Figure 4: TOTP setup screen showing the QR code and a backup secret key

Using the Virtual Keyboard

The virtual keyboard is a security feature designed to protect your TOTP codes from keyloggers and screen-recording malware.

- The keyboard displays digits 0–9 in a **randomized layout** (positions shuffle each time)
- Click the digit buttons to enter your 6-digit TOTP code
- Use the **Back** button to delete the last digit
- Use the **Clear** button to erase all entered digits (this also re-shuffles the layout)
- You can toggle to a physical keyboard input if preferred

The digit positions are randomized each time the virtual keyboard loads or when you press Clear. This means even if someone is watching your screen, they cannot determine your code from button positions alone.

Login & Authentication

Logging In

1. Navigate to `https://192.168.3.44` and click **Login**
2. Enter your registered **email address**
3. Enter your **password**
4. Click **Login**

5. You will be redirected to the TOTP verification page

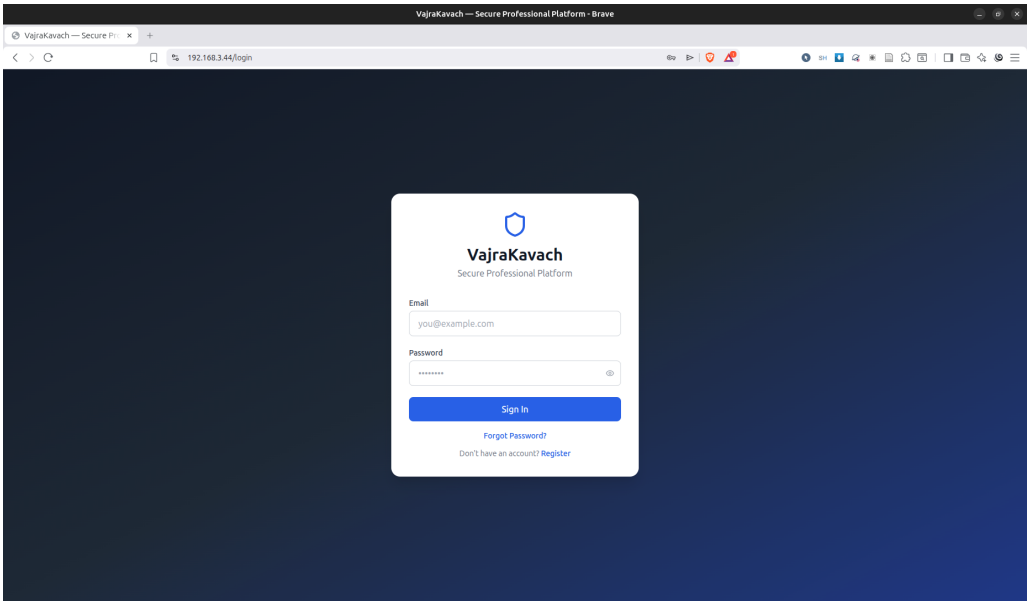


Figure 5: Login page with email and password fields

TOTP Verification

After entering your credentials:

1. Open your authenticator app on your phone
2. Note the current 6-digit code for VajraKavach
3. Enter the code using the **virtual keyboard** on screen
4. Click **Verify**
5. Upon successful verification, you will be redirected to your **Dashboard**

△ Important

Although your authenticator app refreshes the code every 30 seconds, the server accepts the current code **plus one adjacent time-step** (`valid_window=1`), giving each code an effective validity of approximately **90 seconds**. If verification still fails, wait for a completely fresh code and try again.

Account lockout: After **3 failed TOTP attempts within 5 minutes**, your IP address is locked out for **60 minutes**. Subsequent lockouts escalate progressively (×5 each time: 60 min → 300 min → 1500 min, etc.). The lockout is tracked by **IP address**, not email.

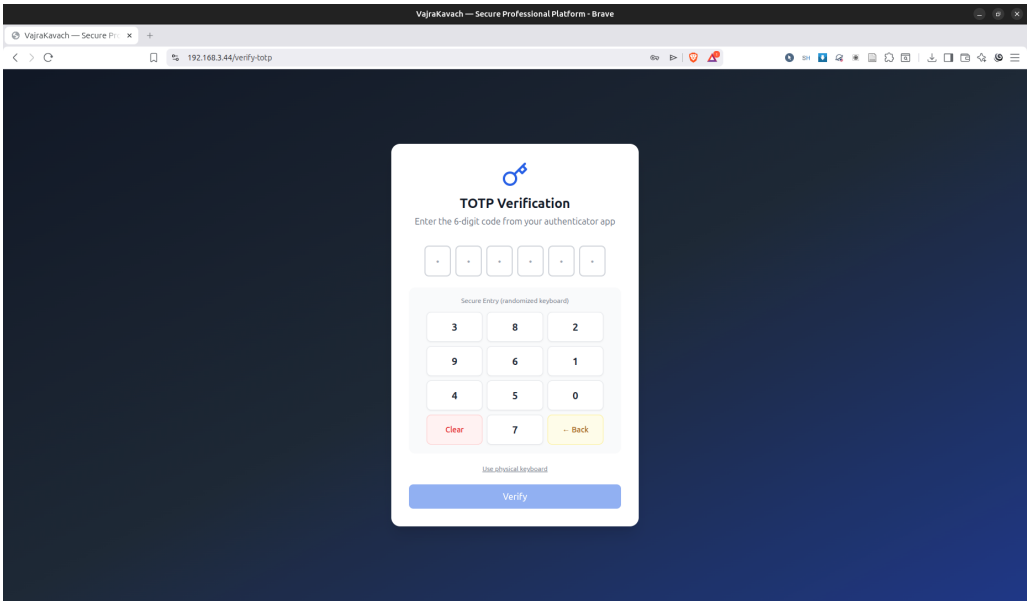


Figure 6: TOTP verification page with the virtual keyboard and a 6-digit input field

Key Import Prompt

If you are logging in from a **new device or browser**, you may see a Key Import prompt. This is because VajraKavach generates unique encryption keys per device for E2EE messaging.

- **Import Key:** If you have a `.vkkey` backup file from your previous device, enter your password and import the file to retain access to your old encrypted messages
- **Dismiss:** If you don't have a backup, dismiss the prompt. A new key pair will be generated, but you won't be able to read messages encrypted with your old key

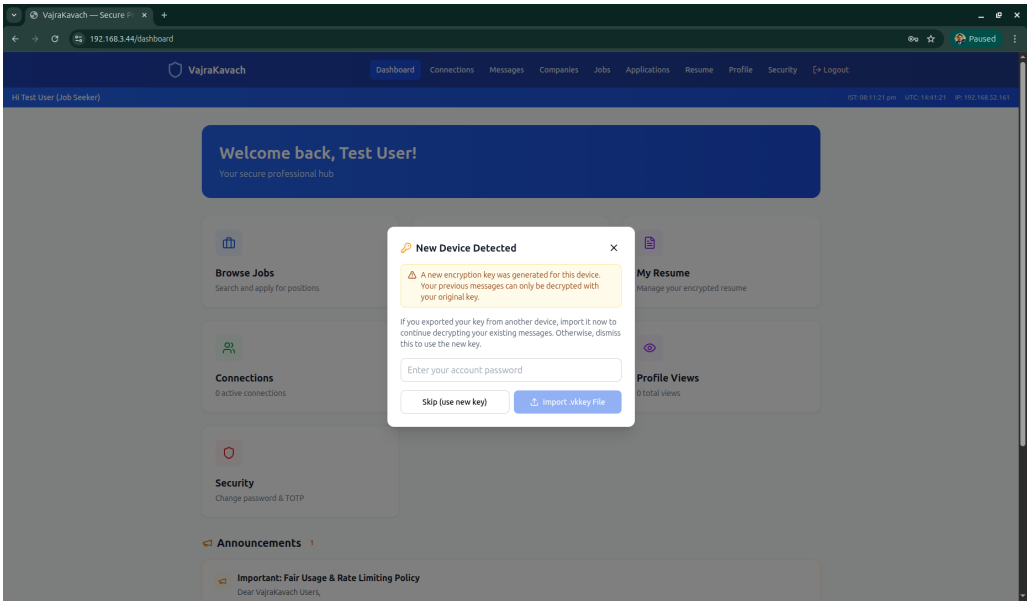


Figure 7: Key import modal dialog asking for password and .vkkey file selection

Forgot Password

If you forget your password, you can reset it using your TOTP:

1. On the login page, click **"Forgot Password?"**
2. Enter your registered **email address**
3. Enter your **new password** (must meet password requirements); a strength indicator is shown
4. Confirm your new password
5. Enter your **TOTP code** using the virtual keyboard
6. Click **Reset Password**

Password reset via TOTP is more secure than email-based reset because it requires possession of your authenticator device. After resetting your password, all existing sessions on other devices are invalidated and you must log in again.

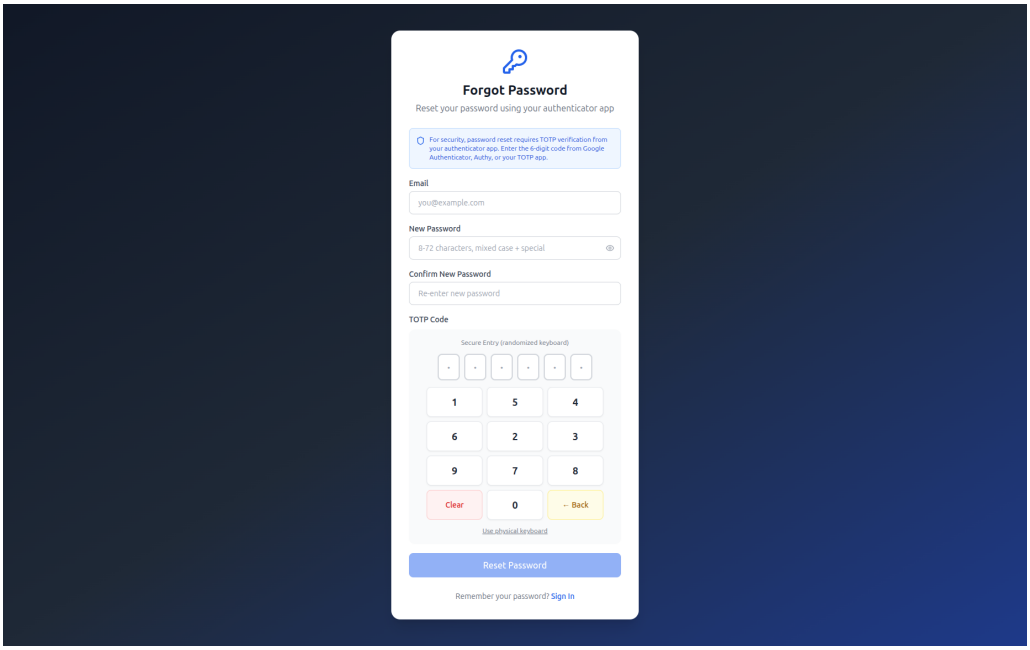


Figure 8: Forgot password page with email, TOTP, and new password fields

Dashboard

After logging in, you are taken to the Dashboard — your central hub for navigating the platform.

User Dashboard

The User (Job Seeker) dashboard displays quick-access cards for:

- **Browse Jobs** — Search and apply for positions
- **My Applications** — Track your submitted applications (shows count)
- **My Resume** — Manage your encrypted resume
- **Connections** — View and manage professional connections (shows count)
- **Messages** — Access your encrypted conversations (shows count)
- **Profile Views** — See how many people viewed your profile
- **Security** — Change password and TOTP settings

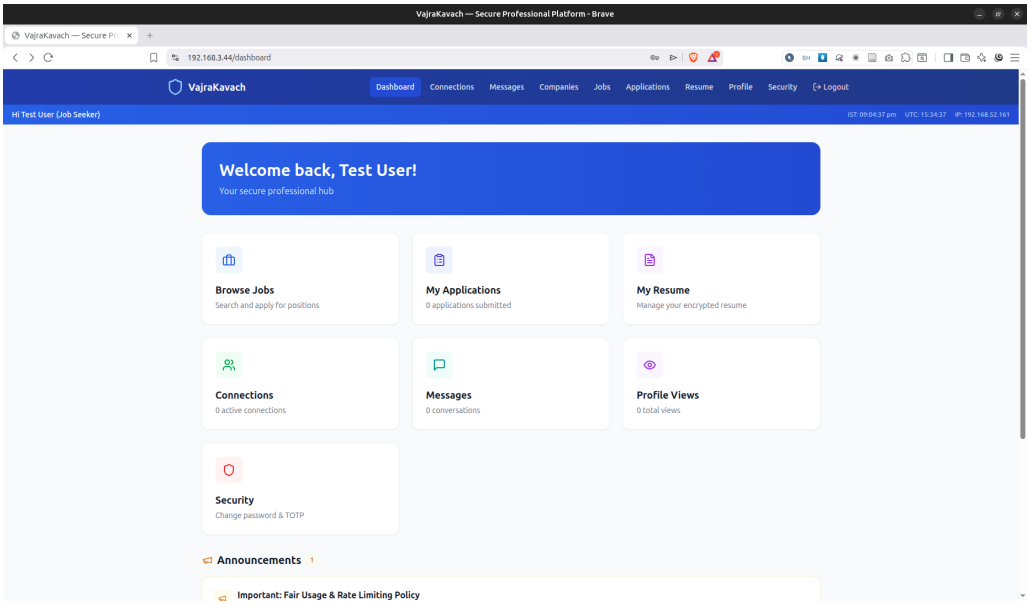


Figure 9: User dashboard showing cards for Browse Jobs, My Applications, My Resume, Connections, Messages, Profile Views, and Security

Recruiter Dashboard

The Recruiter dashboard displays the following cards:

- **My Companies** — View and manage your companies (shows company count)
- **My Job Postings** — View your active job listings (shows listing count)
- **Connections** — View and manage professional connections (shows connection count)
- **Messages** — Access your encrypted conversations (shows conversation count)
- **Profile Views** — See how many people viewed your profile
- **Security** — Change password and TOTP settings

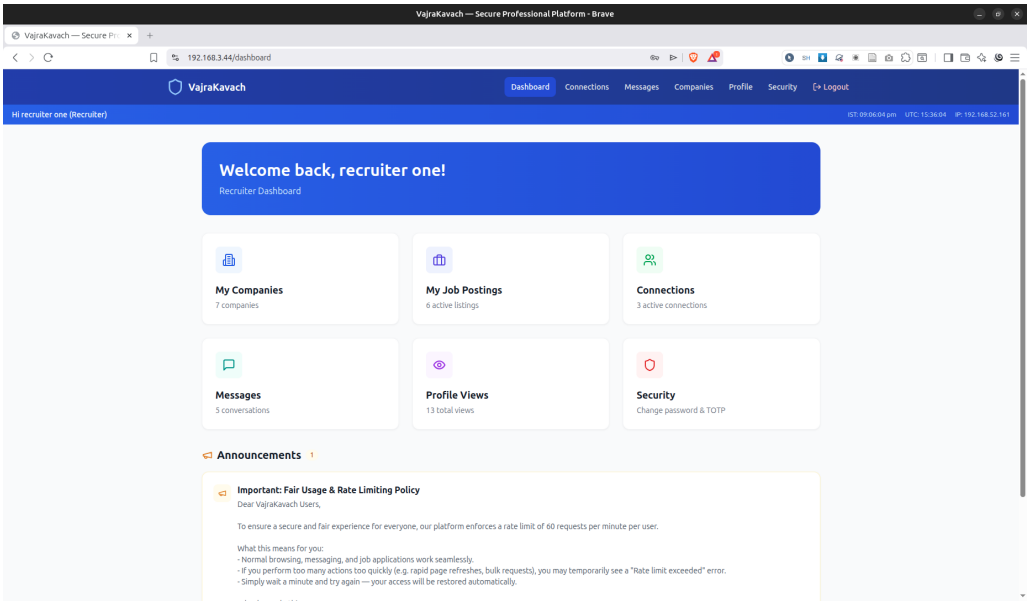


Figure 10: Recruiter dashboard showing cards for My Companies, My Job Postings, Connections, Messages, Profile Views, and Security

Admin Dashboard

The Admin dashboard displays the following cards:

- **Admin Panel** — Access the full admin dashboard to manage platform users (see Section 16)
- **All Jobs** — Monitor all job listings on the platform
- **Companies** — Browse registered companies
- **Security** — Change password and TOTP settings

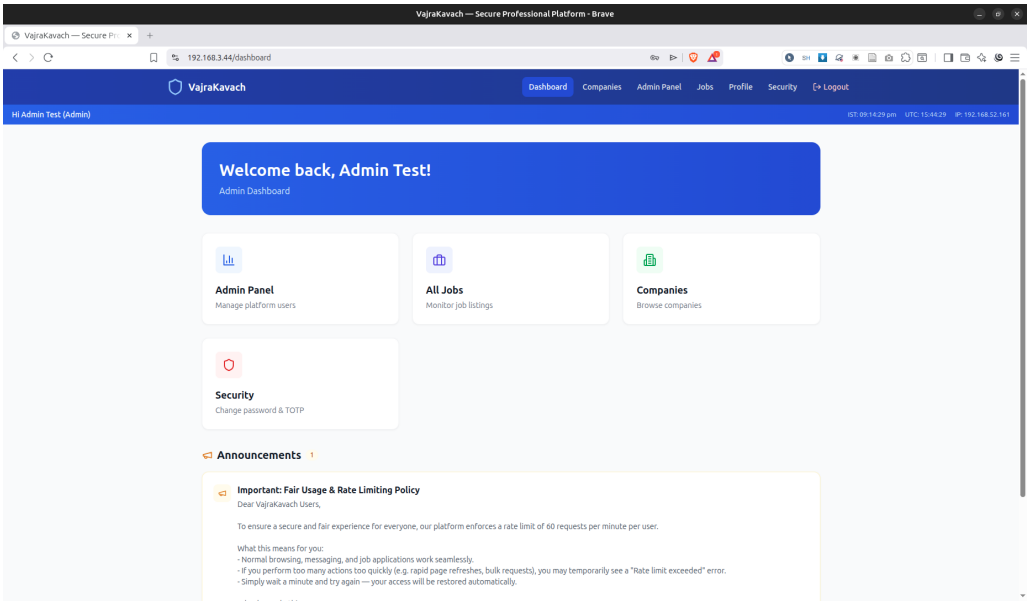


Figure 11: Admin dashboard showing cards for Admin Panel, All Jobs, Companies, and Security

Announcements

If there are any active platform announcements (created by a Platform Admin), they appear below the dashboard cards in a dedicated section:

- Each announcement shows: title, content, sender name, date/time, and a **"Server-side encrypted"** badge
- Announcements are visible to all users regardless of role

Navigation Bar

The top navigation bar is always visible when you are logged in. It provides:

- The **VajraKavach** logo (click to go to Dashboard)
- Role-specific navigation links:
  - **Users:** Dashboard, Connections, Messages, Companies, Jobs, Applications, Resume, Profile, Security
  - **Recruiters:** Dashboard, Connections, Messages, Companies, Profile, Security
  - **Admins:** Dashboard, Companies, Admin Panel, Jobs, Profile, Security
- **Logout** button (right side)

Greeting Bar

Below the navigation bar, a greeting bar displays:

- Your name and role

- Dual clock: **IST** (12-hour format) and **UTC** (24-hour format)
- Your current **IP address** (for session transparency)

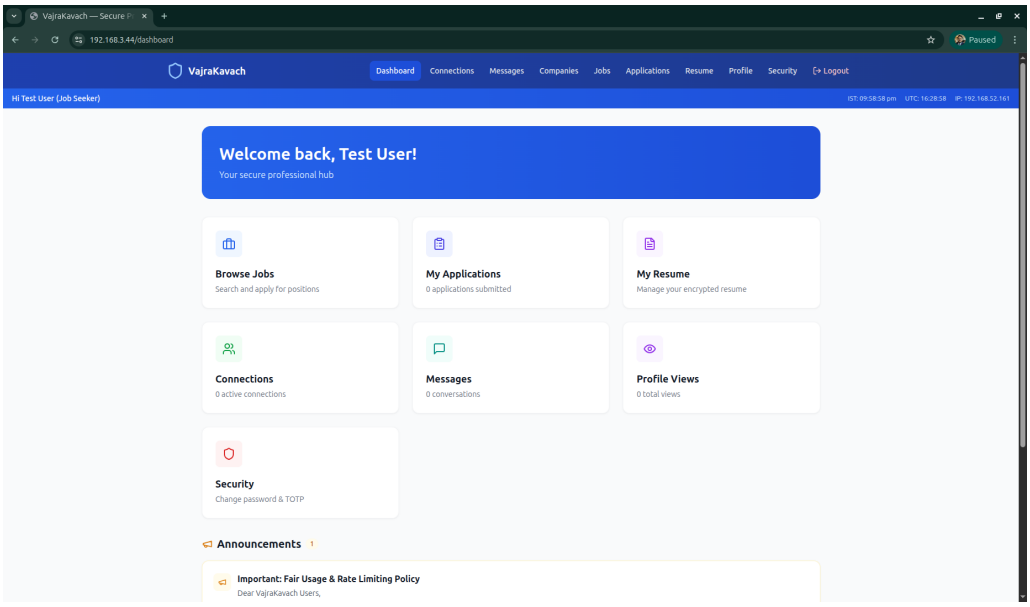


Figure 12: Navigation bar and greeting bar showing user name, role, clocks, and IP address

Profile Management

Viewing Your Profile

Navigate to **Profile** from the navbar or dashboard.

Your profile page displays:

- **Profile view count** in the header (e.g., "12 profile views")
- **Avatar/Profile Picture** with upload button (JPG, PNG, GIF; max 2MB)
- **Basic Information:** Name (editable), Email (read-only), Headline, Bio, Location, Website, Skills (comma-separated)
- **Education** section (list of education entries with Grade/GPA)
- **Experience** section (list of work experiences)
- **Privacy Settings** per field (Public, Connections Only, Private)
- **Encryption Keys** management section (export/import)
- **Profile Viewers** section (viewer list with visibility toggle)

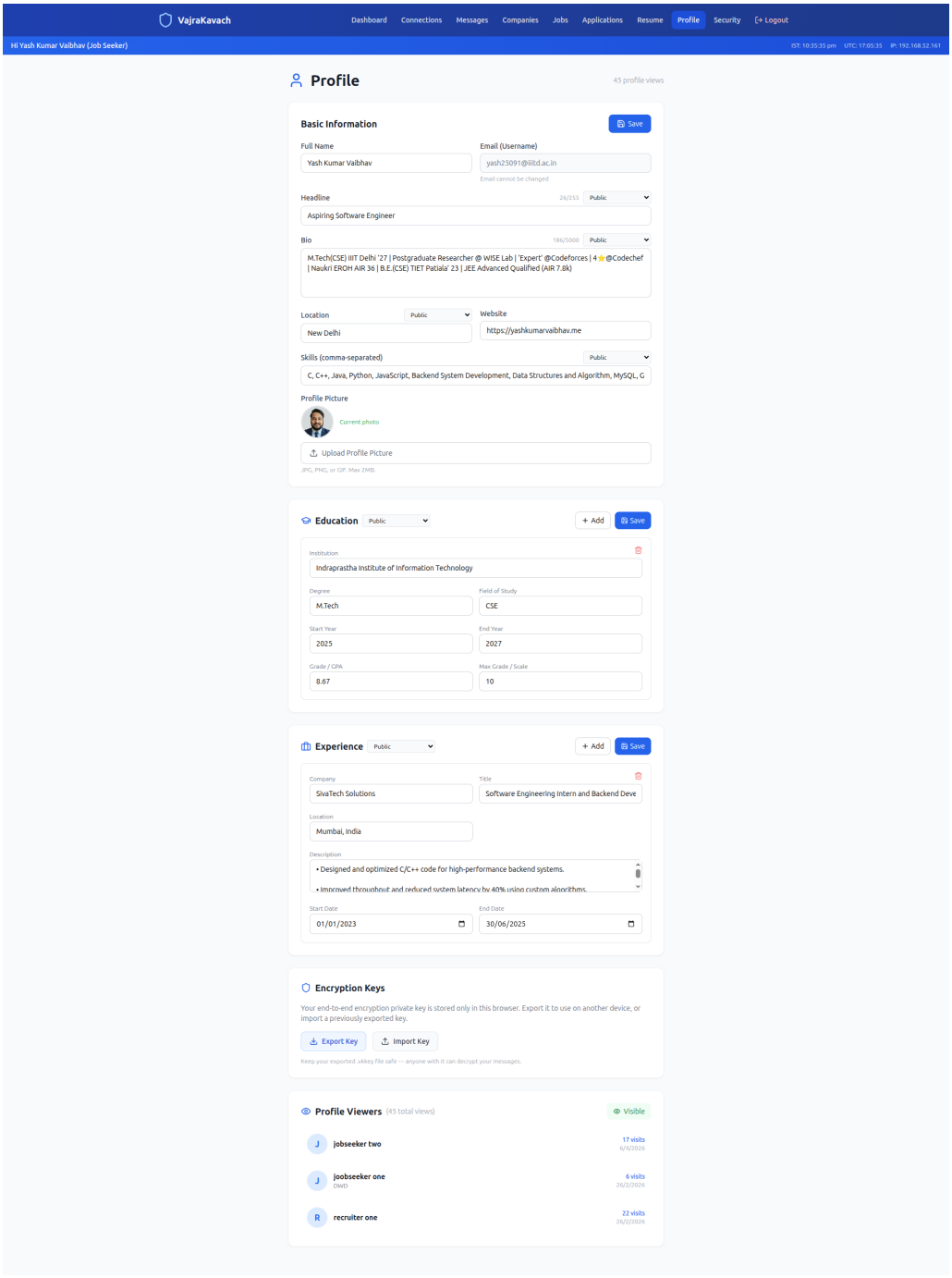


Figure 13: Full profile page showing avatar, basic info, education, experience, and skills sections

Editing Your Profile

All profile fields are edited inline — there is no separate "edit mode." Simply modify any field and click **Save** at the top of the section.

1. In the **Basic Information** section, edit the following fields:
    - **Full Name** — your display name
    - **Headline** — a short professional tagline (e.g., "Software Engineer at Google"), max 255 characters

- **Bio** — a longer description about yourself, max 5000 characters
- **Location** — your city or country
- **Website** — personal or portfolio URL (auto-prepends `https://` if missing)
- **Skills** — comma-separated list (e.g., "Python, React, Machine Learning")

2. Each field (except Name and Email) has a **privacy dropdown** next to it
3. Click **Save** at the top right of the Basic Information card

Your email address cannot be changed after registration. It is displayed as read-only.

Adding Education

1. Scroll to the **Education** section
2. Click **Add Education**
3. Fill in the fields:
   - **Institution** — university or school name
   - **Degree** — e.g., B.Tech, M.Sc.
   - **Field of Study** — e.g., Computer Science
   - **Start Year** and **End Year**
   - **Grade/GPA** and **Max Grade/Scale** (e.g., 9.2 out of 10)
4. Click **Save**
5. To delete an entry, click the **Delete** (trash) icon on the entry

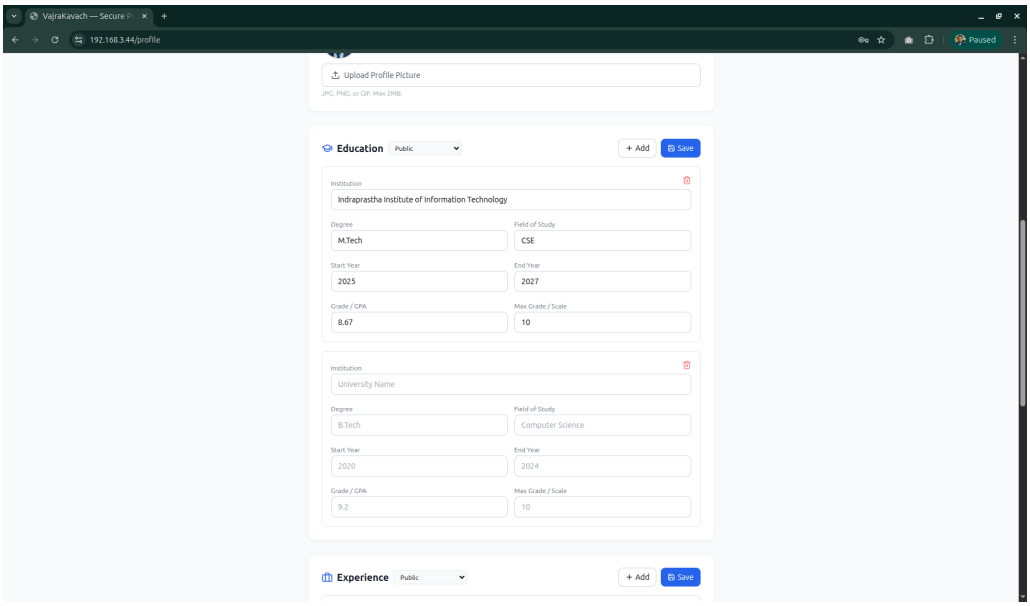


Figure 14: Education section with form fields for institution, degree, field of study, and dates

Adding Experience

1. Scroll to the **Experience** section
2. Click **Add Experience**
3. Fill in the fields:
   - **Company** — company name
   - **Title** — your job title
   - **Location** — city, country
   - **Description** — what you did (max 5000 characters)
   - **Start Date** and **End Date**
4. Click **Save**
5. To delete an entry, click the **Delete** (trash) icon on the entry

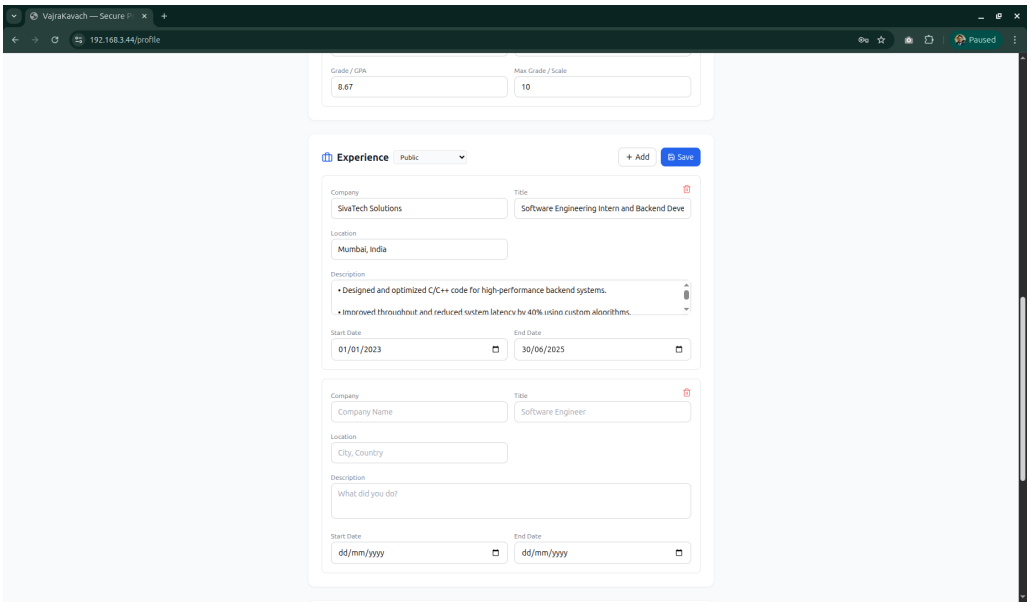


Figure 15: Experience section with form for company, title, location, dates, and description

Privacy Settings

Each profile field can have one of three privacy levels:

Privacy Level	Who Can See
Public	Anyone on the platform
Connections Only	Only your accepted connections
Private	Only you

To change a field's privacy:

1. Click the privacy dropdown next to the field
2. Select the desired privacy level
3. Changes are saved automatically

Profile Picture

1. Click on your current avatar or the camera icon
2. Select an image file from your device
3. The image will be uploaded and displayed as your profile picture

Supported formats include JPG, PNG, and GIF. Maximum file size is 2MB. The image is automatically resized for display.

Profile Viewers

The **Profile Viewers** section at the bottom of your profile shows who has viewed your profile.

- The header shows your total view count (e.g., "12 total views")
- Each viewer entry shows: their name, headline, number of visits, and the date they last viewed
- Click on a viewer's name to visit their profile
- You can toggle viewer list visibility:
  - **Visible** (default) — your viewer list is shown; you can also appear in others' viewer lists
  - **Hidden** — your viewer list is hidden; you also stop appearing in others' viewer lists
- Toggle by clicking the **Visible/Hidden** button (a confirmation prompt will appear)

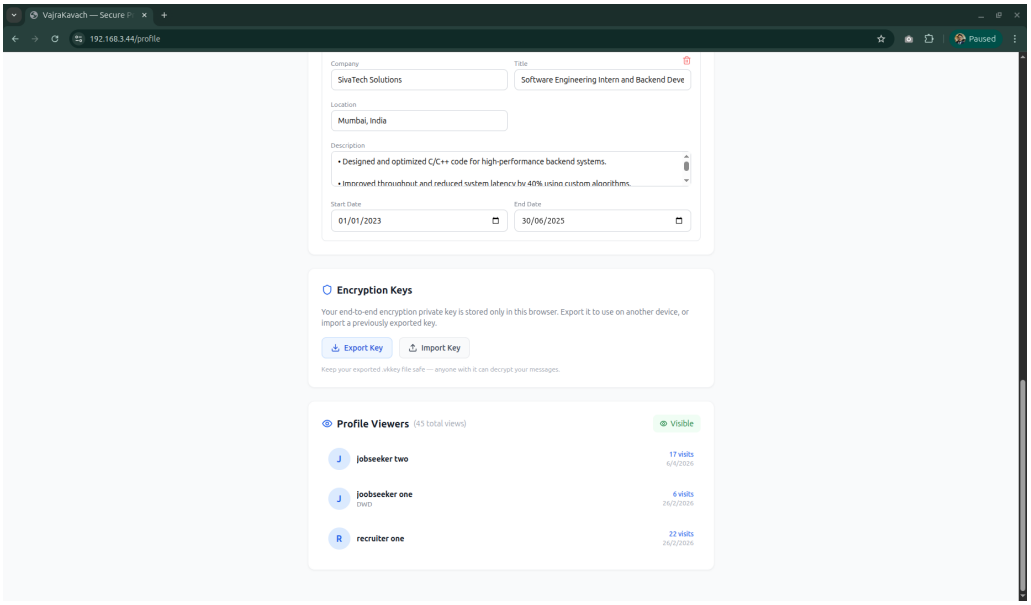


Figure 16: Profile Viewers section showing a list of viewers with visit counts and dates, and the visibility toggle button

Encryption Key Management

VajraKavach uses **End-to-End Encryption (E2EE)** for all private messages. Each user has an **RSA-4096 key pair**:

- **Public Key** — shared with others to encrypt messages to you; stored on the server
- **Private Key** — stored locally on your device (encrypted with your password); **never** sent to the server in plaintext

How Keys are Managed

1. On first login, a key pair is **automatically generated** in your browser
2. The private key is encrypted with your account password and stored in your browser's IndexedDB
3. The public key is sent to the server so others can encrypt messages to you
4. When you close the browser tab, the in-memory (session) key is cleared
5. On next login, the key is decrypted from IndexedDB using your password

Exporting Your Key (Backup)

You must export your key before using the platform on a different device!
Without your key backup, you will lose access to all previously encrypted messages.

1. Navigate to **Profile** or **Security** page

2. Find the **Encryption Keys** section
3. Click **Export Key Pair**
4. A `.vkkey` file will be downloaded to your device
5. Store this file securely (e.g., encrypted USB drive, password manager, or secure cloud storage)

Importing Your Key (New Device)

When you log in on a new device:

1. You may see an automatic **Key Import Prompt**, or navigate to **Profile/Security → Encryption Keys**
2. Click **Import Key Pair**
3. Enter your **account password** (this is verified server-side for security)
4. Select your `.vkkey` backup file
5. Click **Confirm Import**
6. The page will reload and your old messages will become readable again

△ Important

If you dismiss the key import prompt and continue with a new key, your old messages will be **permanently unreadable** on this device. Only import if you have your `.vkkey` file.

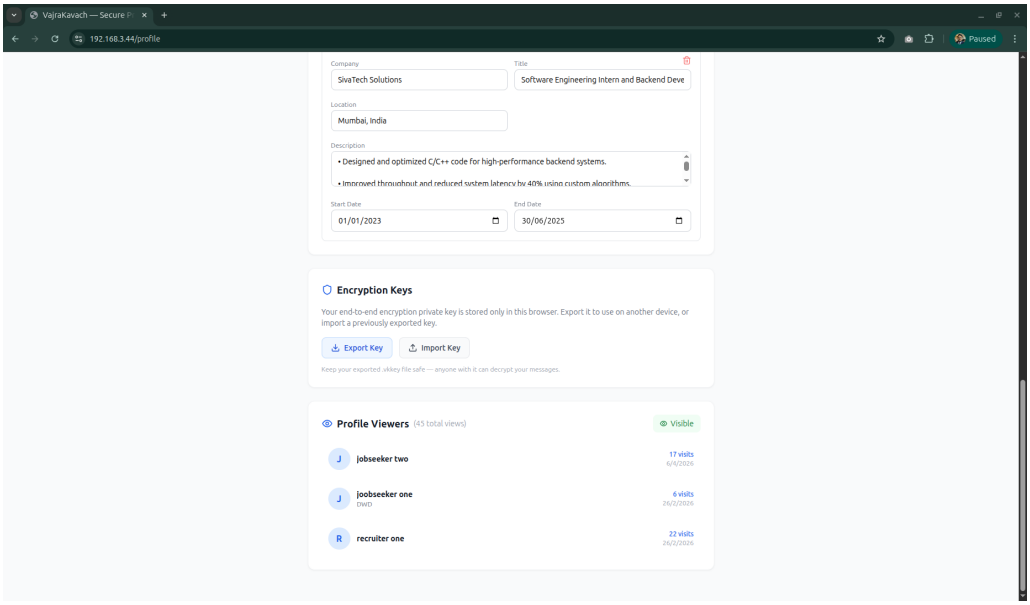


Figure 17: Key import/export buttons in the Encryption Keys section of the profile page

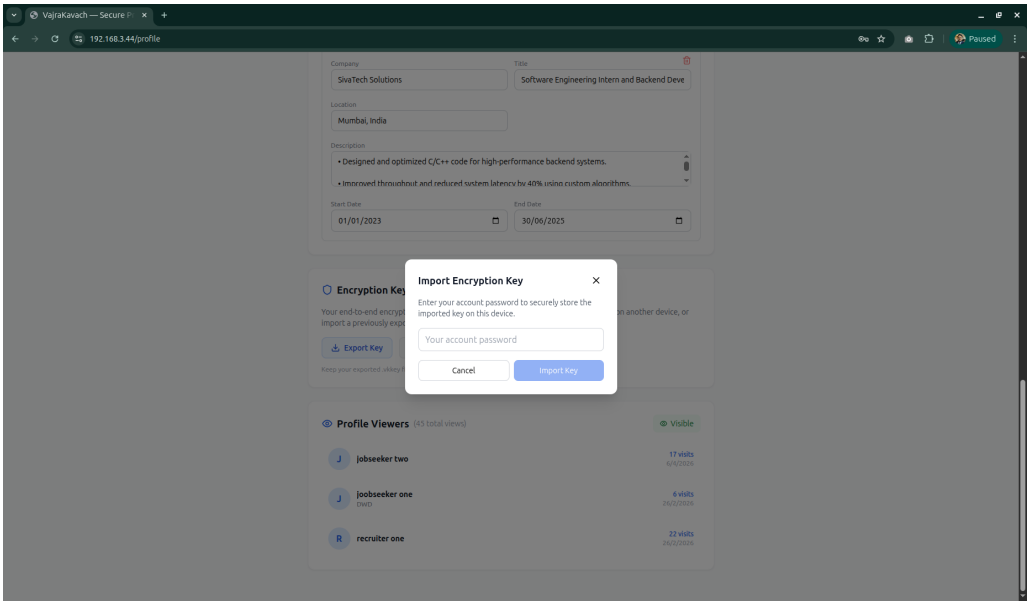


Figure 18: Account password needed for importing Encryption Keys

Resume Management

Uploading a Resume

1. Navigate to **Resume** from the navbar or dashboard
2. Click **Upload Resume**
3. Select a PDF or DOCX file from your device (maximum 10MB)
4. The file is **encrypted automatically** before being stored on the server
5. You will see a confirmation with the filename, upload date, and file size

Your resume is encrypted at rest using Fernet (AES-128-CBC) encryption. Only you, authorized recruiters (to whom you apply), and admins can access it.

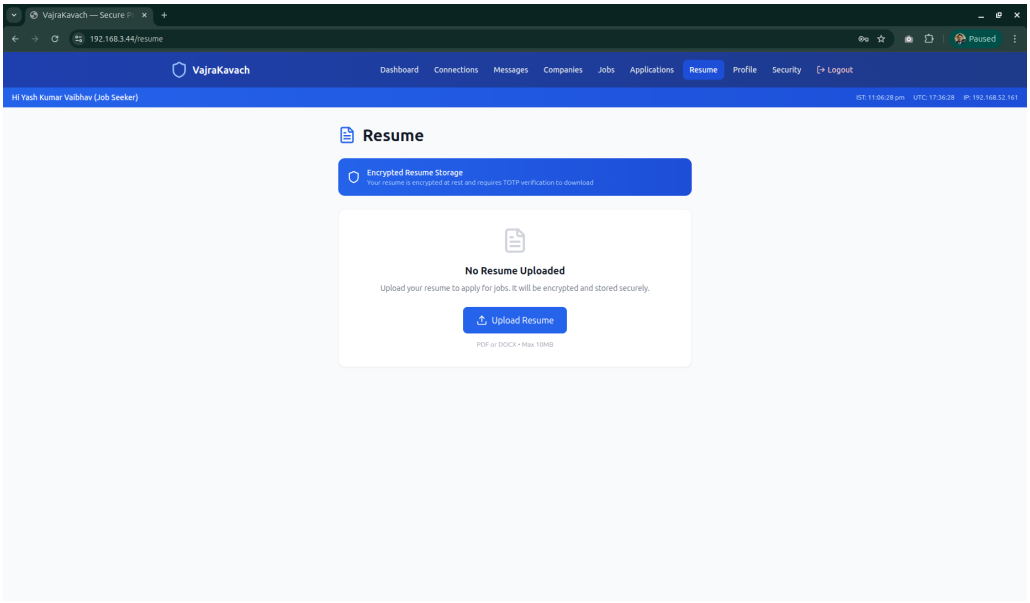


Figure 19: Resume page showing the Upload Resume button and file selector

Downloading Your Resume

Downloading your resume requires TOTP verification as a security measure:

1. Navigate to **Resume**
2. Click **Download Resume**
3. Enter your 6-digit TOTP code using the **virtual keyboard**
4. Click **Verify and Download**
5. The resume will be decrypted server-side and downloaded to your device

△ Important

The TOTP requirement for resume download prevents unauthorized access even if someone gains access to your logged-in session.

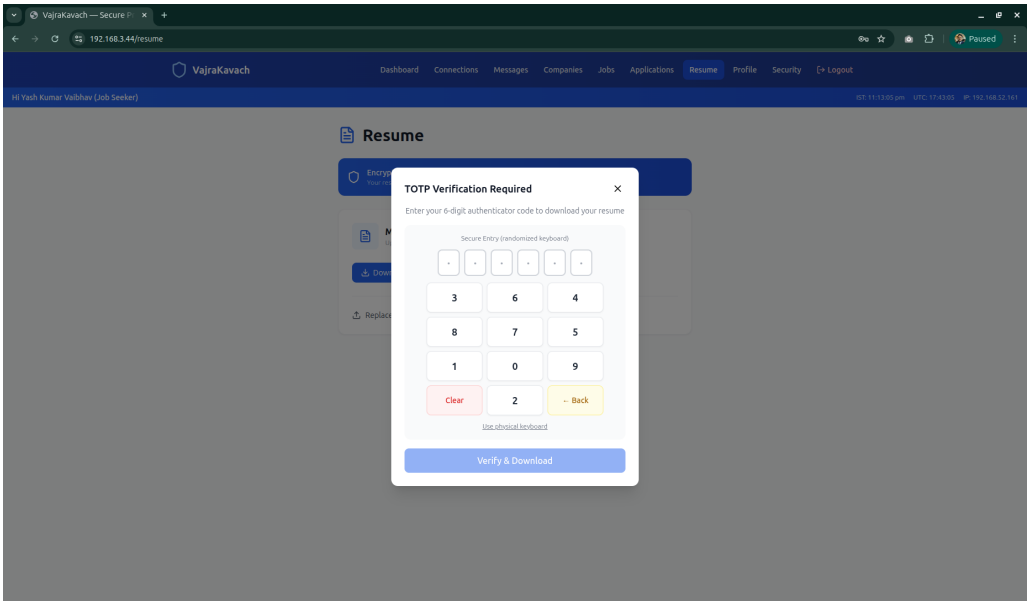


Figure 20: Resume download dialog showing the virtual keyboard for TOTP verification

Updating Your Resume

To update your resume, click **"Replace with new resume"** on the resume card and select a new file. The previous version will be replaced.

Deleting Your Resume

To delete your resume:

1. Click the **Delete** (trash) button on the resume card
2. Enter your **TOTP code** using the virtual keyboard
3. Confirm the deletion

Jobs

Browsing Jobs (All Roles)

1. Navigate to **Jobs** from the navbar
2. Browse the list of active job postings
3. Use the search and filter options:
   - **Keyword search** — search by title, location, or company name
   - **Company filter** — filter by company name
   - **Location filter** — filter by specific city or location text
   - **Skills filter** — filter by required skills (e.g., "Python, React")
   - **Location Type** — All, Remote, On-site, or Hybrid
   - **Job Type** — All, Full Time, Part Time, Internship, or Contract

4. Click **Search** or toggle **Filters** to show/hide the advanced filter panel
5. Click **Clear Filters** to reset all filters

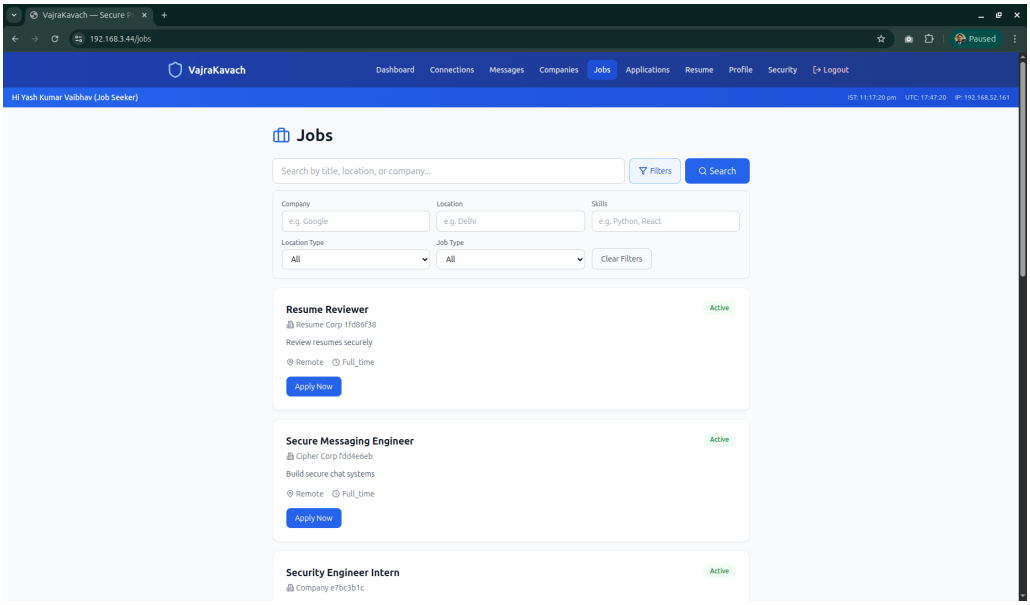


Figure 21: Jobs page showing job listings with search bar and filter options

Applying for a Job (User)

1. Click on a job listing to view details
2. Click the **Apply** button
3. You will be prompted to:
   - Select your **uploaded resume** (if available)
   - Write a **cover letter / cover note**
4. Click **Submit Application**
5. Your application will appear under **Applications** (see Section 10)

Make sure you have uploaded your resume before applying. If you haven't uploaded one yet, navigate to the Resume section first.

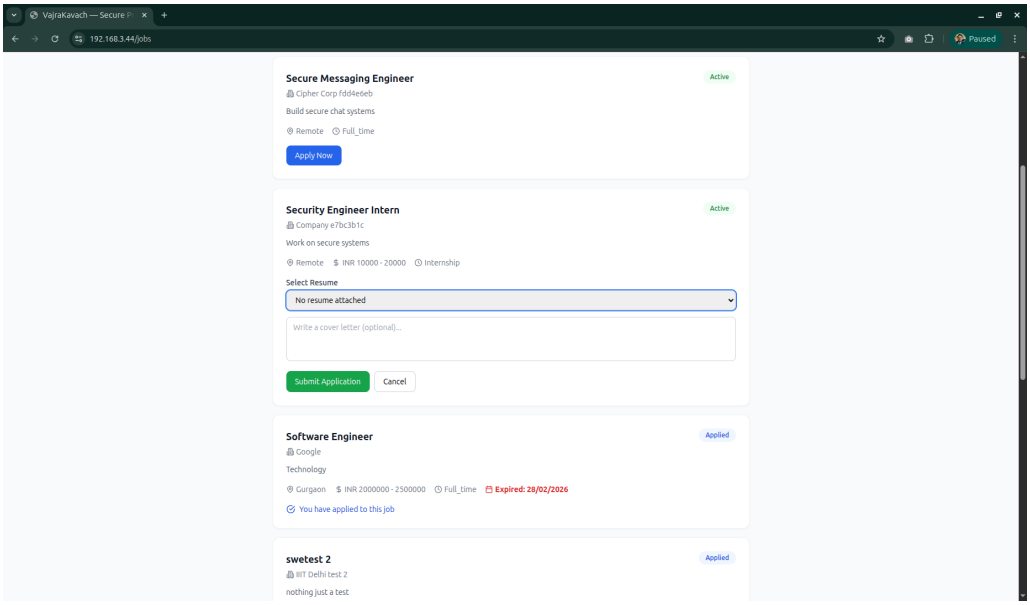


Figure 22: Job application form showing resume selection and cover letter text area

Posting a Job (Recruiter)

Recruiters can post jobs under their company:

1. Navigate to **Companies** → select your company → **Job Postings** tab
2. Click **Post New Job**
3. Fill in the job details:
   - **Job Title** (required)
   - **Description** (required) — detailed job description
   - **Required Skills** — comma-separated skill tags
   - **Location** — city name or leave empty for remote
   - **Currency** — INR, USD, EUR, or GBP
   - **Min Salary** and **Max Salary** (optional)
   - **Job Type** — Full Time, Part Time, Contract, or Internship
   - **Location Type** — On-site, Remote, or Hybrid
   - **Application Deadline** (date picker, must be today or later)
4. Click **Post Job**
5. The job will be immediately visible to all users in the Jobs section

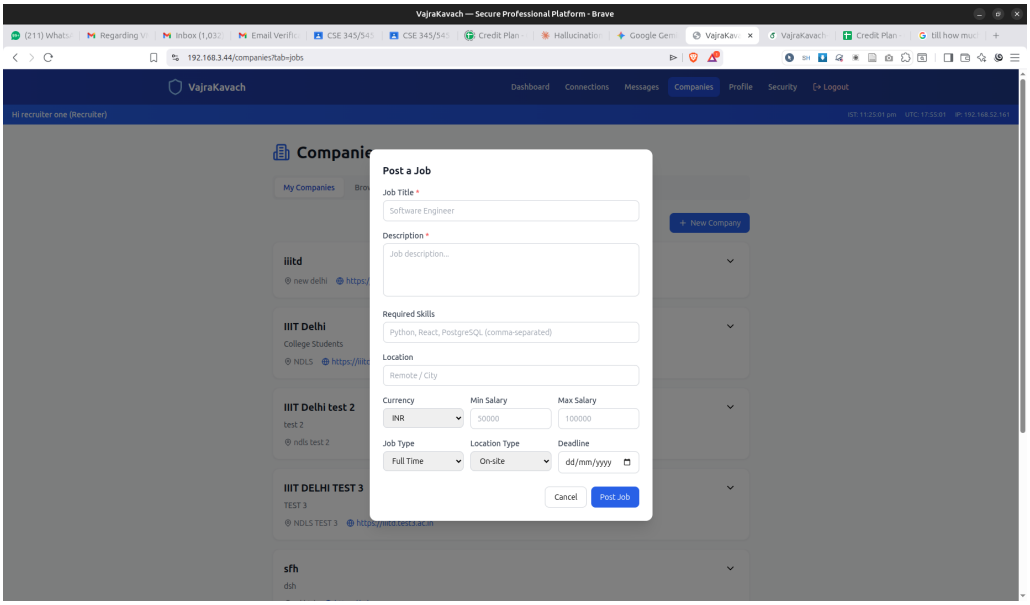


Figure 23: Job posting form with fields for title, description, skills, location, type, salary, and deadline

Applications

Tracking Your Applications (User)

1. Navigate to **Applications** from the navbar
2. View a list of all jobs you have applied to
3. Each application shows:
   - **Job title** and **company name**
   - **Application date**
   - **Current status:** Applied, Reviewed, Interviewed, Rejected, or Offer
4. Click on an application for more details
5. You can **withdraw** an application if the status is still "Applied"

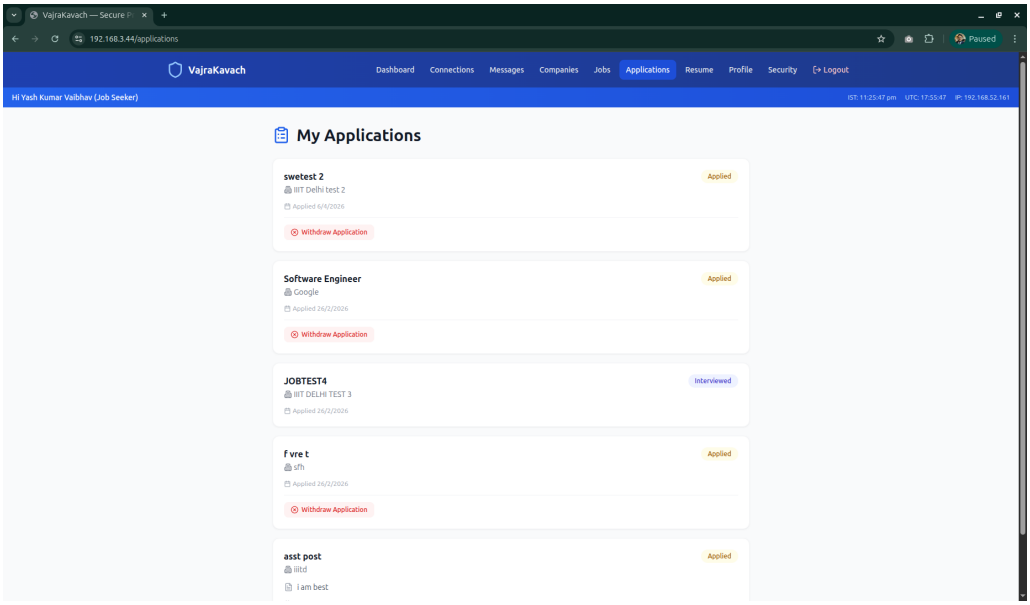


Figure 24: Applications page showing a list of job applications with status badges

Managing Applicants (Recruiter)

Recruiters can manage applicants from the **Companies** section:

1. Navigate to **Companies** → select your company
2. Go to the **Job Postings** tab
3. Click on a job posting to view applicants
4. For each applicant, you can:
   - **View their profile** (name, headline, skills)
   - **Download their resume** (requires TOTP verification via virtual keyboard)
   - **Update application status** using the status dropdown:
     - Applied → Reviewed → Interviewed → Offer / Rejected
   - **Add internal notes** about the candidate (click the Notes icon)
   - **Message the candidate** directly
   - **View Match Score** — auto-compares the candidate's resume skills against the job requirements, showing matched/missing skills and a percentage score

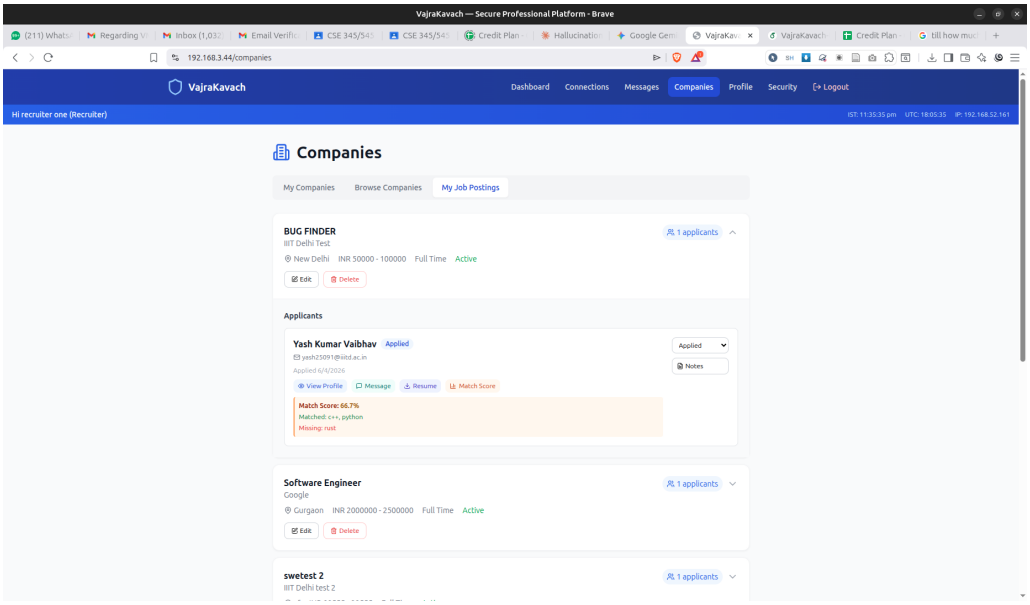


Figure 25: Recruiter view showing applicants for a job with status dropdowns and action buttons

Companies

Creating a Company Page (Recruiter)

1. Navigate to **Companies** from the navbar
2. Click the **My Companies** tab
3. Click **Create Company**
4. Fill in the company details:
   - **Company Name** (required)
   - **Description** (required) — about the company
   - **Location** (required) — company headquarters
   - **Website** (required)
   - **Industry** (optional) — e.g., Technology, Finance
   - **Size** (optional) — e.g., 100–500 employees
5. Click **Create**
6. You are now the administrator of this company page

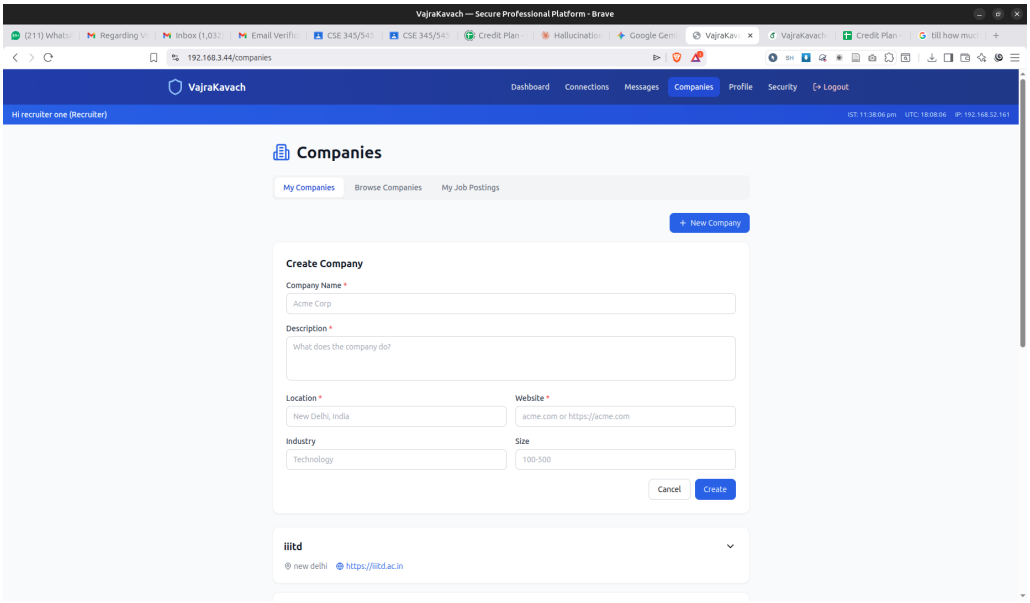


Figure 26: Create Company form with name, description, location, and website fields

Browsing Companies (All Roles)

1. Navigate to **Companies**
2. Click the **Browse Companies** tab
3. Use the **search bar** to search by name or location
4. Use the **Industry** dropdown to filter by industry
5. Click the **Sort** button to toggle alphabetical sort (A→Z / Z→A)
6. Browse company cards showing name, industry badge, description, location, and website

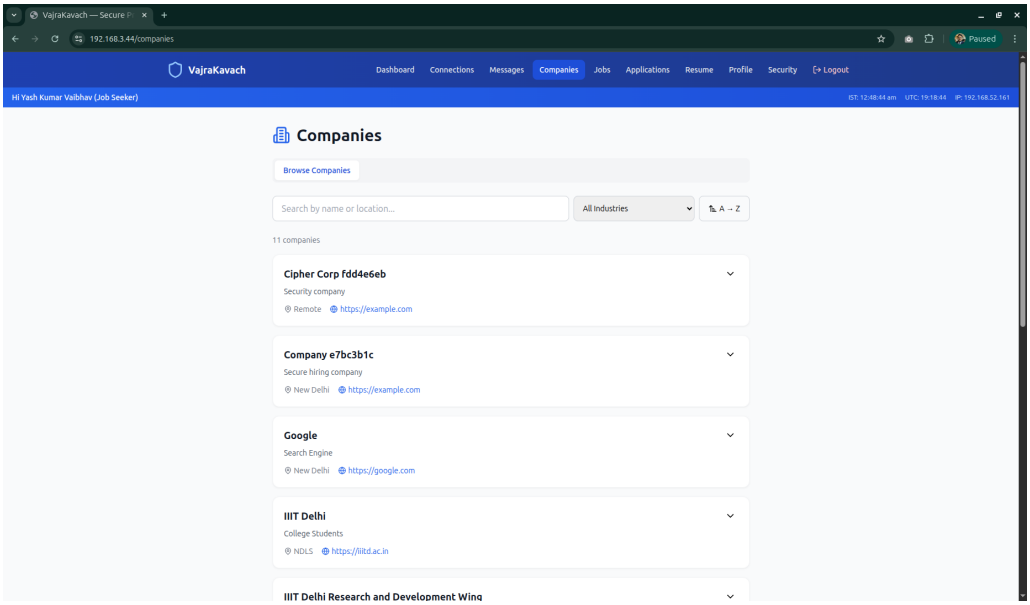


Figure 27: Browse Companies tab showing a grid of company cards with names and descriptions

Managing Job Postings (Recruiter)

1. Navigate to **Companies**
2. Click the **My Job Postings** tab
3. This shows **all jobs posted by you** across all your companies
4. Here you can:
   - View all your job listings with company name, status, and applicant count
   - **Edit** a job listing (click the Edit icon to modify title, description, skills, salary, etc.)
   - **Delete** a job listing (with confirmation dialog)
   - Toggle job status (active/closed via the Active checkbox in the edit form)
   - View and manage applicants — click the expand arrow to see all applicants for a job (see Section 10)
5. To post a new job, go to the **My Companies** tab, expand your company card, and click **"Post a Job"** (see Section 9.3)

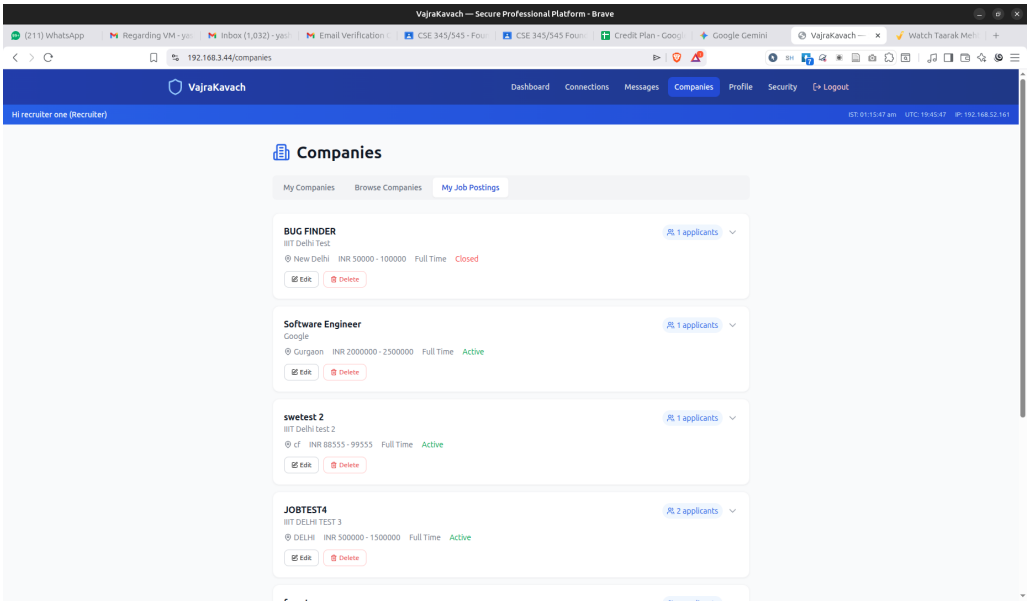


Figure 28: My Job Postings tab showing all job listings posted by the recruiter

Connections

Finding People

1. Navigate to **Connections** from the navbar
2. Click the **Find People** tab
3. Browse platform users or search by name
4. Click **Connect** next to a user to send a connection request

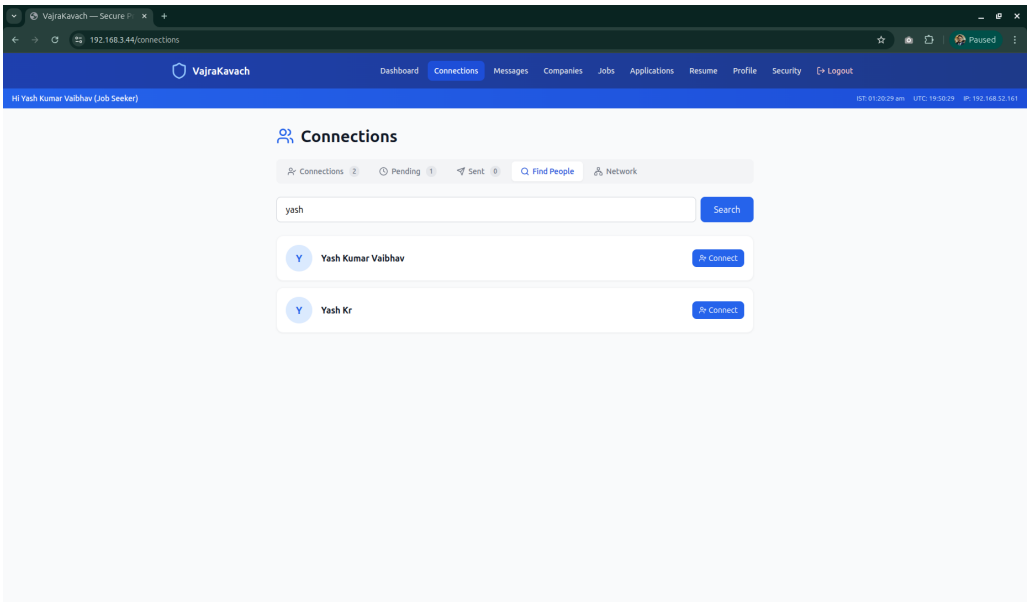


Figure 29: Find People tab showing user cards with Connect buttons

Managing Connection Requests

Connections are managed through multiple tabs:

Tab	Description
Connections	View all accepted connections. Click a user to view their profile.
Pending	Incoming connection requests from others. Accept or decline.
Sent Requests	Outgoing requests you have sent. Cancel if needed.
Find People	Discover new users and send connection requests.
Network	Visualize your connection graph interactively.

Connection Graph

The **Network** tab provides an interactive force-directed graph visualization of your professional network:

- Your profile appears as a larger node at the center labeled "You"
- Direct connections are shown as linked nodes (sized by their connection count)
- **Solid lines** represent direct connections; **dashed green lines** indicate mutual connections
- **Green badges** on edges show the number of mutual connections
- Hover over a node to see the user's name, role (Recruiter/Job Seeker), and mutual count in a tooltip below the graph
- Click on a node to navigate to that user's profile
- A **legend** at the bottom explains the node colors and line styles

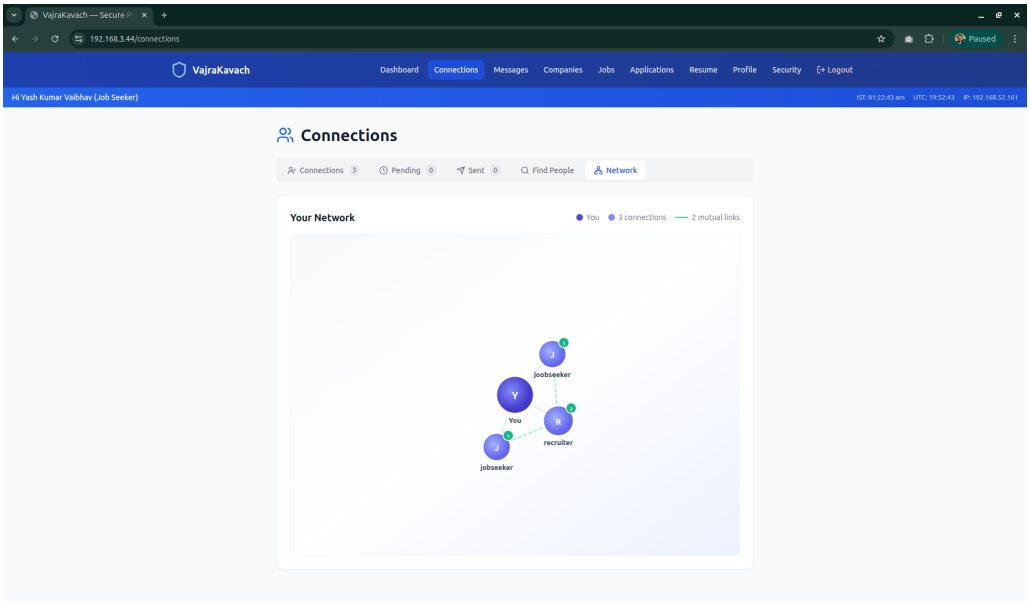


Figure 30: Interactive network graph visualization showing the user at center with connected nodes

Messaging (End-to-End Encrypted)

Overview

All messages on VajraKavach are **End-to-End Encrypted (E2EE)**. This means:

- Messages are encrypted in your browser before being sent
- The server stores only **ciphertext** — it cannot read your messages
- Only the sender and recipient(s) can decrypt and read the messages

For E2EE to work, both you and the recipient must have valid encryption keys. If you see a warning saying "keys unavailable," export/import your keys from the Profile or Security section.

Sending a Direct Message

1. Navigate to **Messages** from the navbar
2. Click the + button next to "Conversations" to open the New Conversation dialog
3. The dialog shows your connections who don't have an existing conversation yet — click on a user to start a DM
4. Alternatively, select an existing conversation from the sidebar
5. Type your message in the input field at the bottom

6. Click **Send** (or press Enter)
7. The message is encrypted with the recipient's public key and your own public key (so both of you can read it)

You can also start a conversation from other pages (e.g., clicking the Message icon on a connection's profile or on an applicant card). This redirects to the Messages page with the conversation pre-selected.

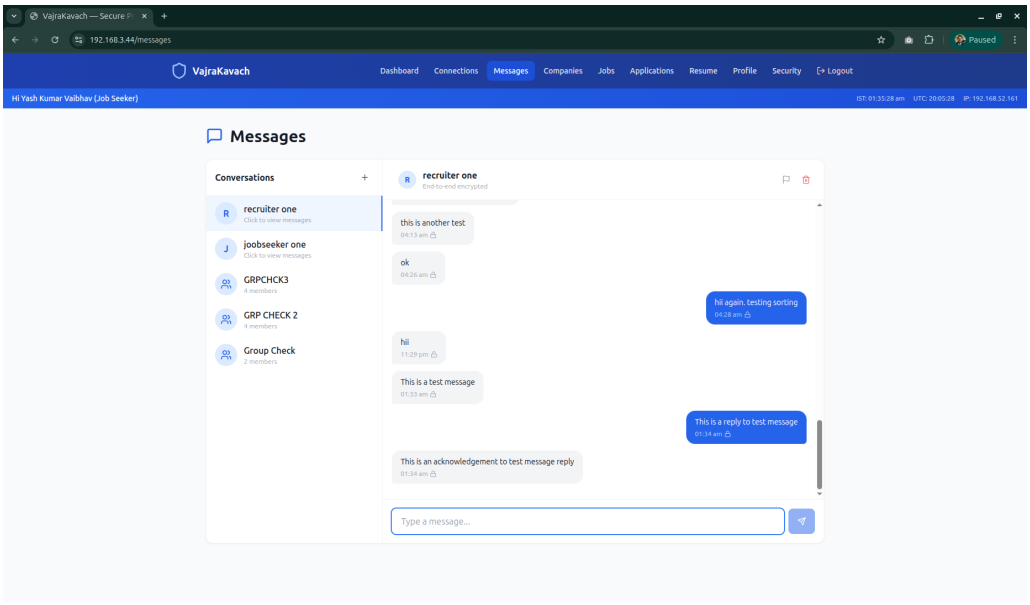


Figure 31: Direct message conversation showing encrypted messages between two users

Group Messages

1. In the Messages section, click the + button next to "Conversations"
2. Click **Create Group Chat**
3. Enter a **group name**
4. Select members from your connections (checkboxes)
5. Click **Create Group**
6. Messages in groups are individually encrypted for each member

Group Management

The group creator has additional controls:

- **Info** — click the Info button in the chat header to see group details and members
- **Add Members** — add new connections to the group

- **Remove Members** — remove a member from the group
- **Delete Group** — archive the group (no new messages can be sent)

Non-creator members can:

- **Leave** the group (you can still view old messages but cannot send new ones)
- **Delete** the group from your view

Group messages are encrypted separately for each member using their individual public keys. This ensures true E2EE even in group chats.

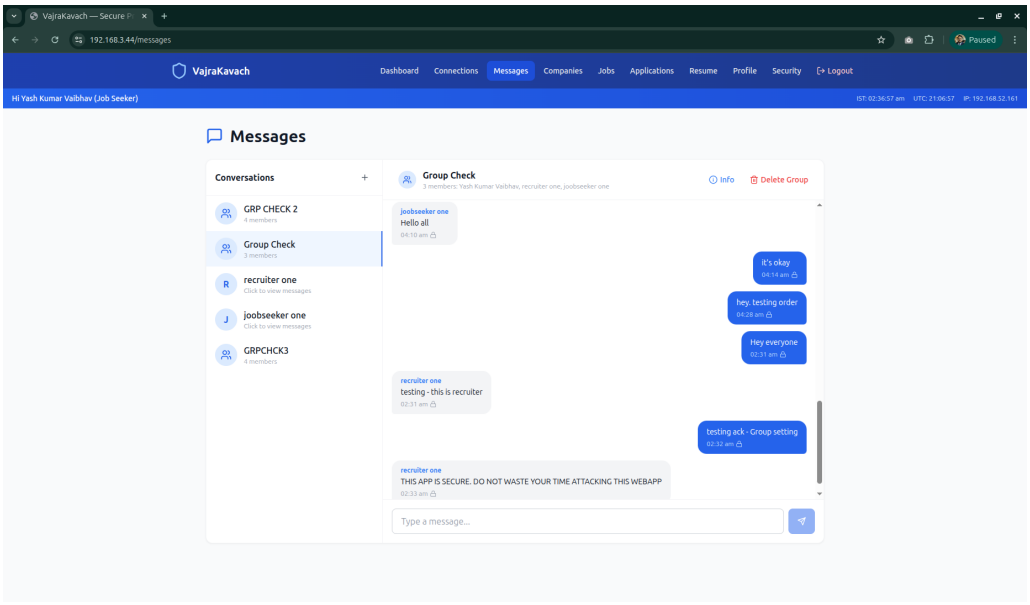


Figure 32: Group chat showing multiple participants and encrypted messages

Message Encryption Indicators

In the message view:

- Encrypted messages are displayed as normal readable text (decrypted in your browser)
- If a message cannot be decrypted (e.g., wrong key), it will display as *"[Encrypted — unable to decrypt]"* in yellow italic
- A **lock icon** next to the timestamp indicates E2EE is active
- For direct messages, the chat header displays **"End-to-end encrypted"**

Deleting Messages

- Hover over any message to reveal a **Delete** (trash) icon
- For your own messages, you can choose:
  - **Delete for me** — removes the message from your view only
  - **Delete for everyone** — removes the message for all participants
- For others' messages, only **Delete for me** is available
- Deleted messages show as *"This message was deleted"* with a dashed border

Messaging from Other Pages

You can also start a message from:

- **ViewProfile page** — click the **Message** button on a connected user's profile

Viewing Other Users' Profiles

1. Click on any user's name elsewhere on the platform (connections, applicants, admin panel, etc.)
2. You will see their **public profile** with:
   - Name, headline, location, email, website (subject to their privacy settings)
   - About / Bio section
   - Skills (as tag pills)
   - Education and experience (if visible)
3. Available actions (depending on connection status):
   - **Connect** — send a connection request (if not connected)
   - **Message** — start an encrypted conversation (if already connected)
   - If a request is pending, you will see a **"Request Pending"** badge

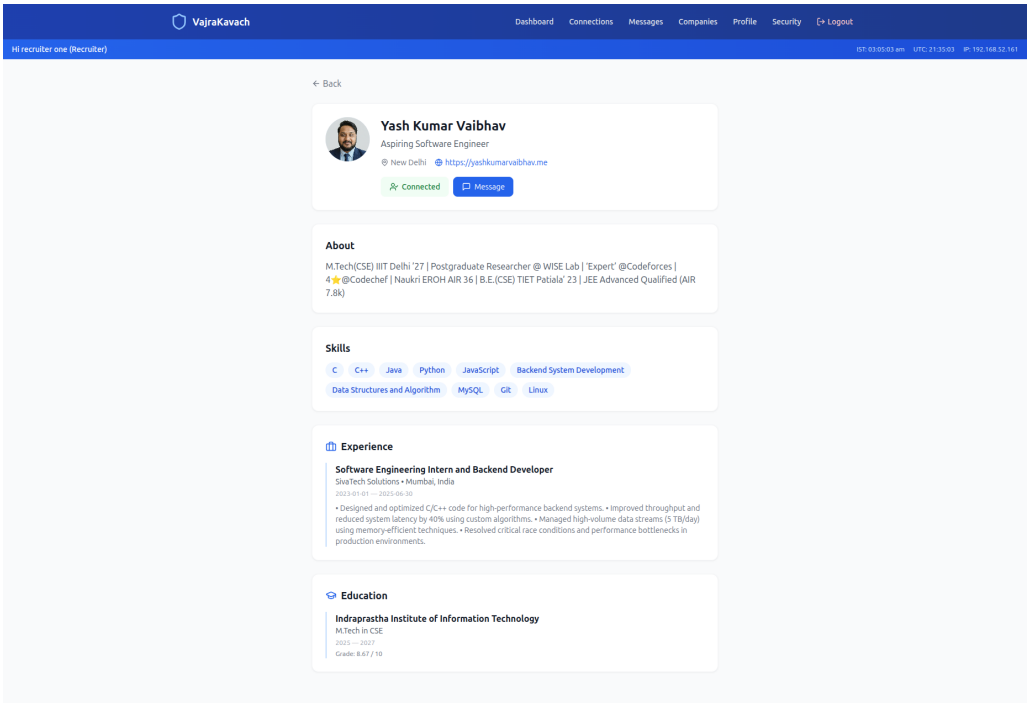


Figure 33: Read-only profile view of another user with Connect and Message buttons

Security Settings

Navigate to **Security** from the navbar to access security-related account actions.

At the top, a **Security Status** banner shows three green indicators:

- **TOTP Active** — your two-factor authentication is enabled
- **E2E Encryption** — end-to-end encryption is active
- **PKI Verified** — your public key infrastructure is set up

Changing Your Password

1. On the Security page, find the **Change Password** section
2. Enter your **current password**
3. Enter your **new password** (8–72 characters, must include uppercase, lowercase, digit, and special character); a **password strength bar** with a checklist is shown
4. Confirm your new password
5. Enter your **TOTP code** using the virtual keyboard
6. Click **Change Password**

△ Important

After changing your password, you will need to re-import your encryption key (`.vkkey` file) since the key is encrypted with your password.

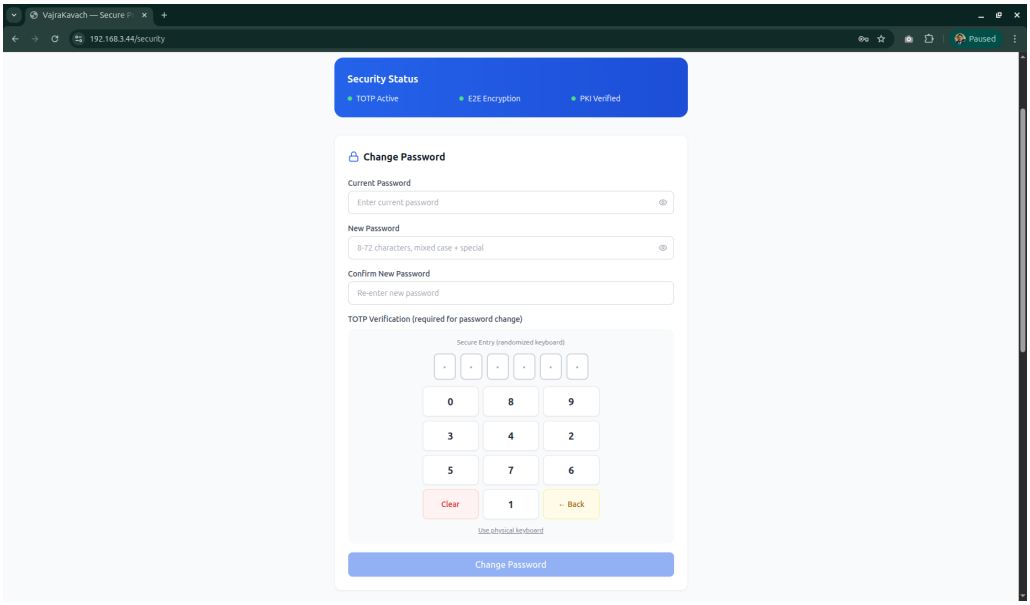


Figure 34: Change Password form with current password, new password, confirm, and TOTP fields

Deleting Your Account

Account deletion is **permanent and irreversible**. All your data, messages, applications, and uploaded files will be permanently removed.

1. On the Security page, find the **Delete Account** section (in the red-bordered "Danger Zone")
2. Type the confirmation phrase `DELETE MY ACCOUNT` exactly as shown
3. Enter your **TOTP code** using the virtual keyboard
4. Click **Delete My Account Permanently**
5. You will be logged out and your account will be permanently deleted

Account Information

The Security page also displays your **Account Information** in a read-only card:

- **Email** — your registered email address
- **Role** — your current role (User, Recruiter, Admin)
- **TOTP** — verification status ("Verified")
- **Status** — account status ("Active")

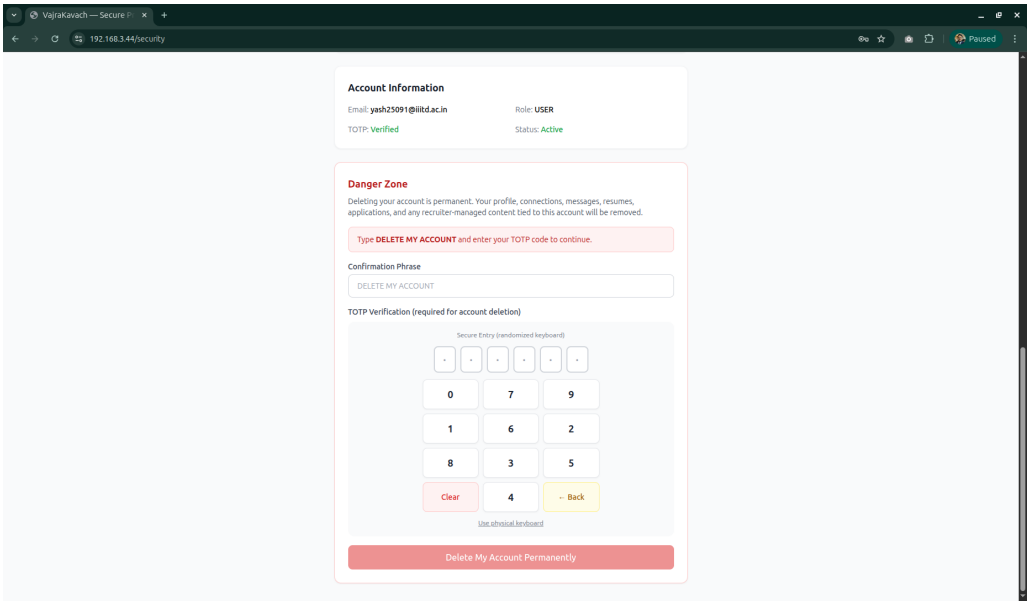


Figure 35: Delete Account section with confirmation text input and TOTP virtual keyboard

Admin Dashboard

The Admin Dashboard is available to users with the **Admin** role. Access it via the **Admin Panel** link in the navbar.

TOTP Verification Gate

Before accessing the Admin Dashboard, you must verify your identity with TOTP:

1. Enter your 6-digit TOTP code using the virtual keyboard
2. Click **Verify & Access**
3. Once verified, the full admin dashboard loads

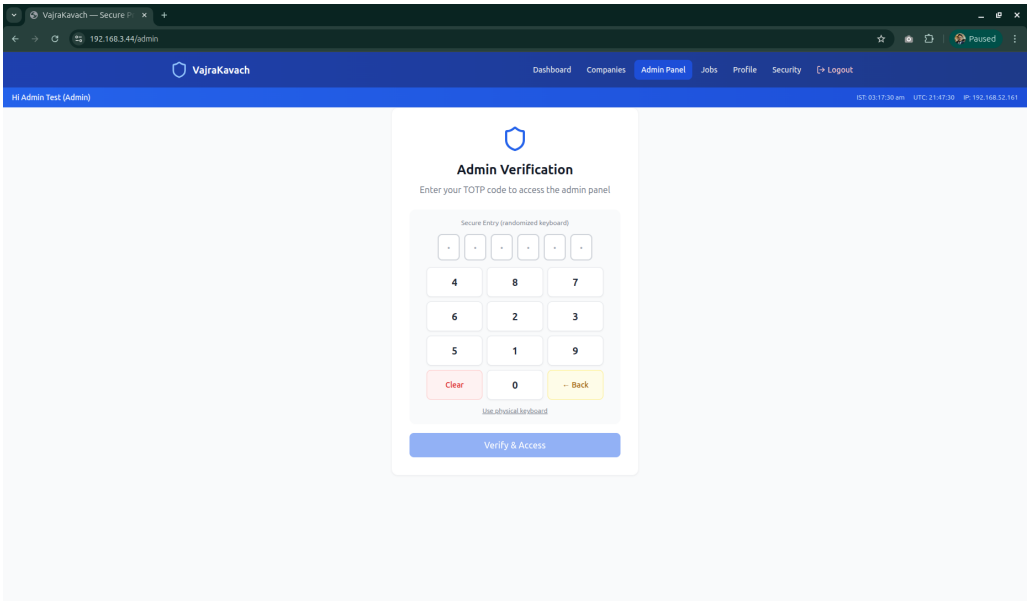


Figure 36: Admin TOTP verification screen with virtual keyboard before accessing the dashboard

△ Important

The Admin role you select during registration provides **view-only access** to the admin dashboard. You can see all data (users, jobs, resumes, audit logs, reports, blockchain) but **cannot take any actions** such as suspending users, delisting jobs, creating announcements, mining blocks, or resolving reports. Those write actions are reserved for the Platform Admin (super admin), which is a role assigned internally and not available through registration. The dashboard header will display **"Admin Dashboard (View Only)"** to indicate this.

Overview Tab

The Overview tab provides aggregate platform statistics displayed as colored cards:

- **Total Users** — total registered users on the platform
- **Active Users** — users with active accounts
- **Suspended** — users who have been suspended
- **Total Jobs** — total job postings
- **Audit Logs** — total audit log entries

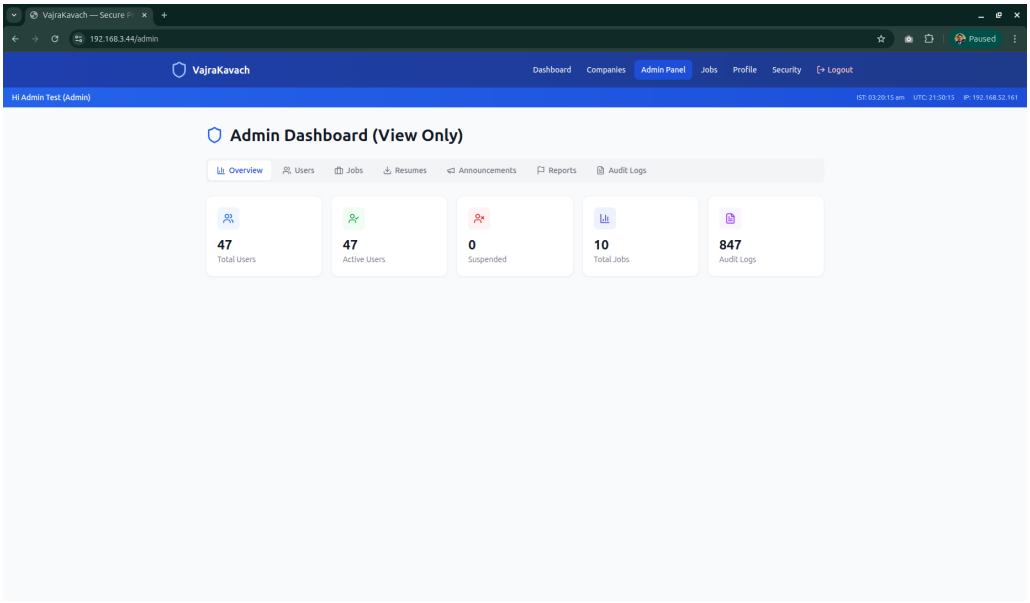


Figure 37: Admin dashboard overview showing aggregate statistics as cards and charts

User Management

1. Click the **Users** tab
2. View a paginated table of all registered users with:
   - Name, email, role, status (active/suspended)
   - Last login IP address
   - Registration date
3. **Search** for users by name or email

As an Admin (view-only), you can browse and search all users but cannot suspend, activate, delete, or change roles. Action buttons (Suspend, Delete, Promote) are only visible to Platform Admins.

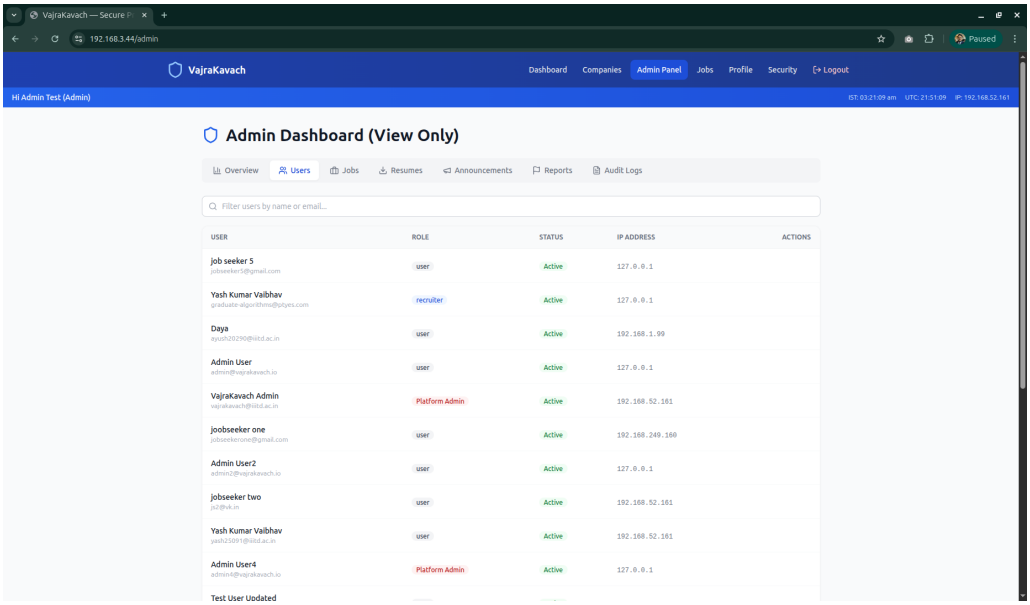


Figure 38: Admin Users tab showing a table of users with columns for name, email, role, status, and actions

Job Moderation

1. Click the **Jobs** tab
2. View all job postings across the platform
3. Each listing shows: title, company name, location, type, active status, and creation date

As an Admin (view-only), you can browse all job listings but cannot delist or modify them. The delist toggle button is only visible to Platform Admins.

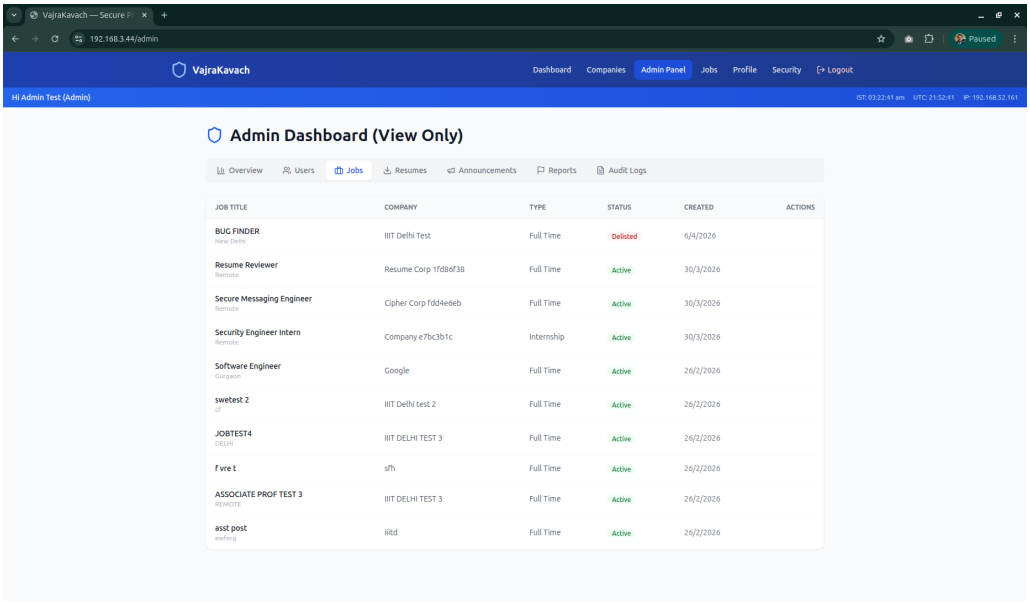


Figure 39: Admin Jobs tab showing job listings with delist action buttons

Resume Overview

1. Click the **Resumes** tab
2. View metadata of all uploaded resumes (file name, upload date, owner)
3. Admins can download resumes by verifying their TOTP token; the file is decrypted server-side upon authorised download

Announcements

1. Click the **Announcements** tab
2. View all platform-wide announcements

As an Admin (view-only), you can view existing announcements but cannot create or delete them. The announcement creation form is only visible to Platform Admins.

User Reports

1. Click the **Reports** tab
2. View all abuse reports submitted by users
3. Each report includes:
   - Reporter information
   - Reported user
   - Category (Spam, Harassment, Fraud, etc.)

- Description
- Timestamp

4. Take appropriate moderation action based on reports

As an Admin (view-only), you can view all reports but cannot change their status (mark as reviewed/resolved). Status update buttons are only visible to Platform Admins.

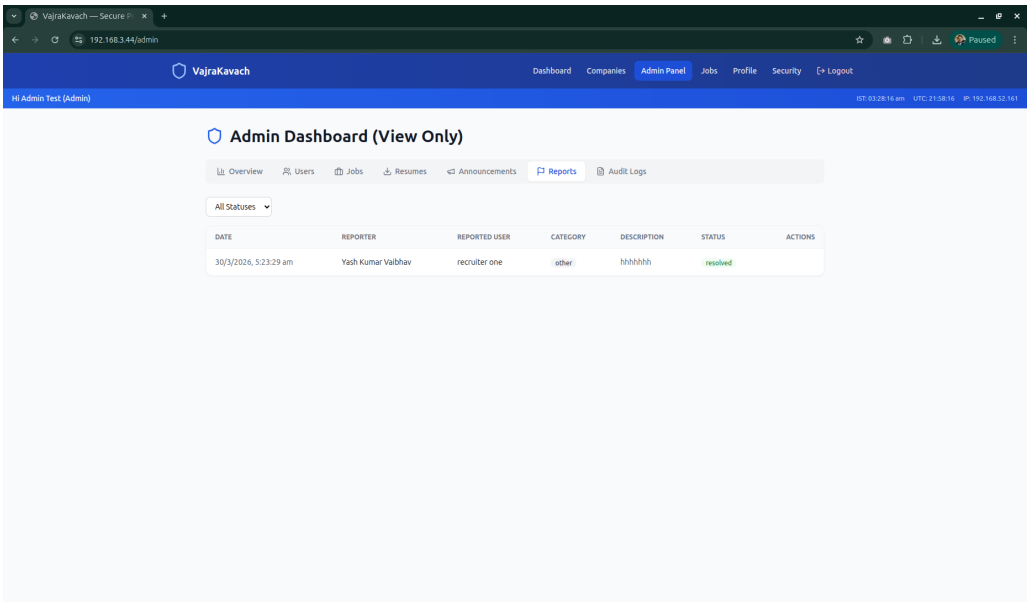


Figure 40: Admin Reports tab showing a list of user reports with categories and descriptions

Audit Logs

1. Click the **Audit Logs** tab
2. View a paginated history of all security-relevant actions:
   - User logins, registrations, and TOTP verifications
   - Profile updates and account changes
   - Resume uploads, downloads, and deletions
   - Message sends and group creation
   - Connection requests
   - Job/company creation, updates, and deletions
   - Application submissions and status changes
   - Admin actions (user suspend/activate, job moderation)
   - Blockchain mining events
   - E2EE key uploads

3. Each log entry shows a **Valid** or **Tampered** integrity badge, automatically verified via hash chaining
4. Use the **Previous / Next** buttons to paginate through entries

Audit logs use **hash chaining** — each log entry includes a SHA-256 hash of the previous entry. This makes tampering detectable, as modifying any entry would break the chain.

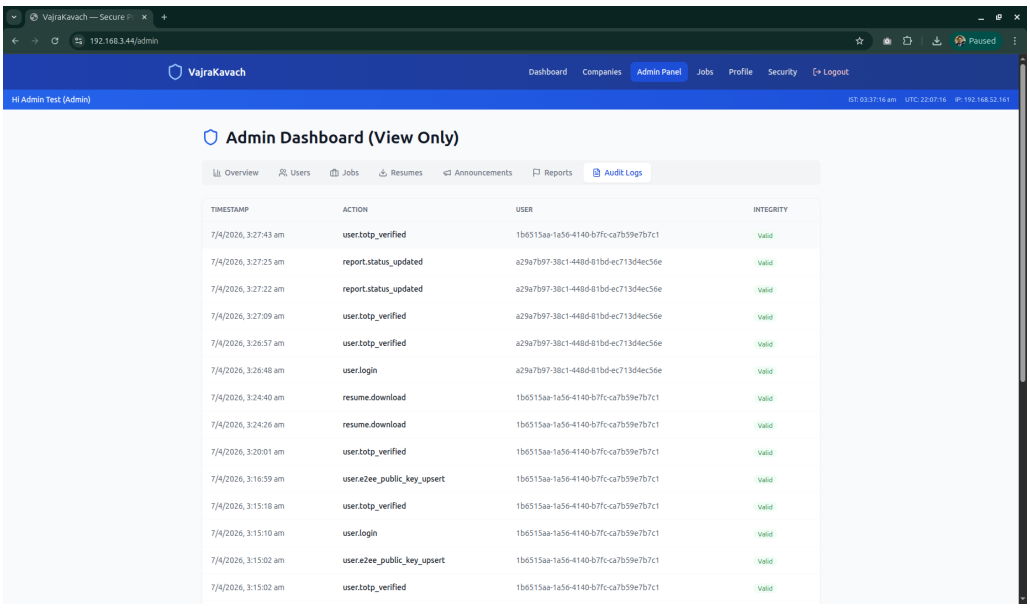


Figure 41: Audit Logs tab showing log entries with timestamps, actions, users, and hash chain verification button

Blockchain Verification

The platform includes optional blockchain-based tamper-evidence:

1. In the Audit Logs section, click **Blockchain** sub-tab
2. Click **Verify Blockchain** to verify the integrity of all mined blocks
3. View individual blocks with their Merkle roots and hash pointers

As an Admin (view-only), you can view blockchain blocks and verify their integrity, but cannot mine new blocks. The **Mine Block** button is only visible to Platform Admins.

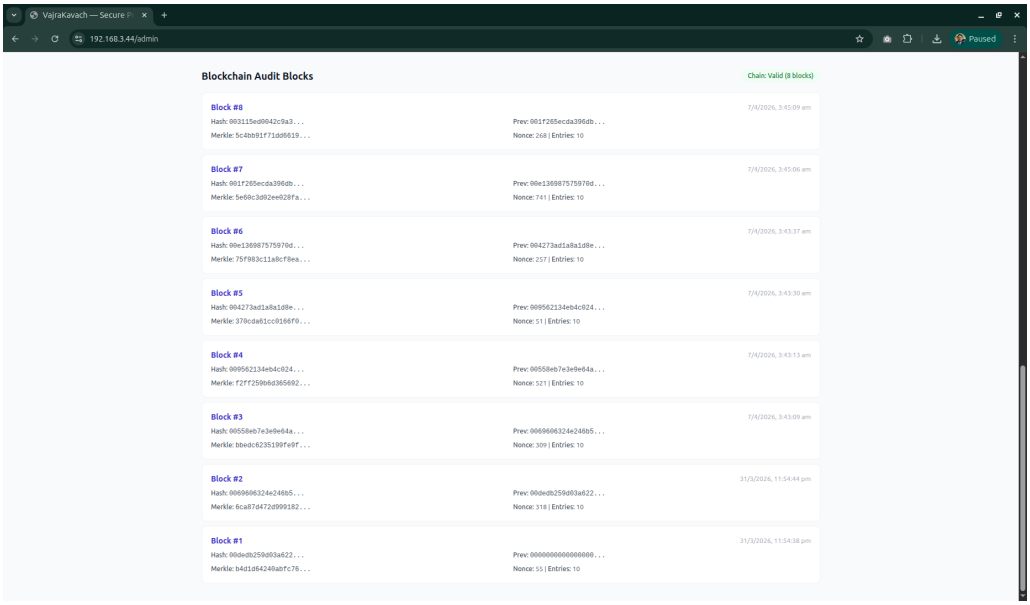


Figure 42: Blockchain verification page showing mined blocks with hashes and verification status

Reporting Users

If you encounter inappropriate behavior, you can report a user:

1. In a **direct message conversation**, click the **Report** button (flag icon) in the chat header
2. In the report dialog:
   - Select a **Category**: Spam, Harassment, Fraud/Scam, Inappropriate Content, Impersonation, or Other
   - Provide a **Description** of the issue (5–2000 characters)
3. Click **Submit Report**
4. The report is sent to platform admins for review

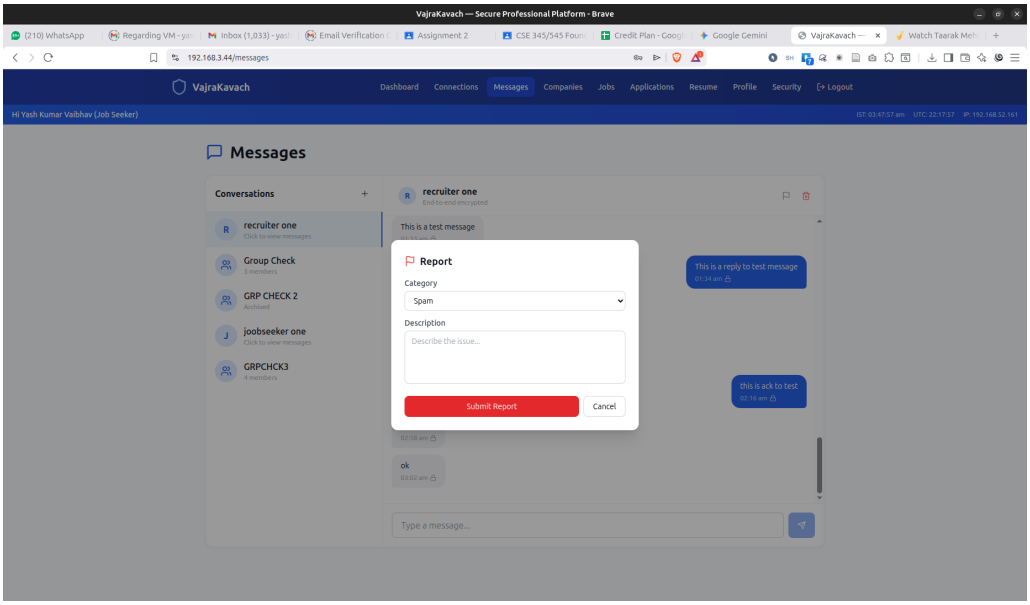


Figure 43: Report dialog with category dropdown and description textarea

Troubleshooting

Issue	Solution
Browser shows "Your connection is not private"	Install the rootCA.pem certificate (see Section 2.2)
TOTP code not working	Ensure your phone's clock is synced (use automatic time). TOTP is time-based and requires accurate clocks. Wait for a fresh code.
Cannot read old messages on new device	Import your `.vkkey` file from your old device. Without it, old messages encrypted with your old key pair cannot be decrypted.
"Keys unavailable" warning in Messages	Go to Profile → Encryption Keys → Export/Import your key pair.
Resume download fails	Ensure you enter the correct TOTP code. If the issue persists, try uploading a new resume.
Account locked after failed TOTP attempts	After 3 failed attempts within 5 minutes your IP is locked out for 60 minutes. Subsequent lockouts escalate progressively (×5 each time: 300 min, 1500 min, etc.). Wait for the lockout to expire and ensure you are using the correct TOTP code.
Cannot connect to `192.168.3.44`	Ensure you are on the same LAN. The platform is not accessible from the public internet.

Page not loading after login	Clear your browser cache and cookies, then log in again.
Forgot password & lost TOTP	Contact a platform administrator — account recovery without TOTP requires manual intervention.

Security Best Practices

Follow these recommendations to keep your account secure:

1. **Use a strong, unique password** — do not reuse passwords from other sites
2. **Back up your TOTP secret** — use an authenticator app that supports cloud backup, or save the setup QR code securely
3. **Export your encryption key** — save your `.vkkey` file immediately after first login; store it securely
4. **Never share your `.vkkey` file** — anyone with this file and your password can read your messages
5. **Log out when done** — especially on shared or public computers
6. **Verify the URL** — always ensure you are on `https://192.168.3.44` with the padlock icon
7. **Keep your authenticator app updated** — update your phone's OS and authenticator app regularly
8. **Report suspicious activity** — if you notice unauthorized access or suspicious users, use the report feature
9. **Check your IP address** — the greeting bar shows your session IP; verify it matches your expected location

Quick Reference Card

Action	Navigation Path	Notes
Register	Landing Page → Get Started	Requires TOTP setup
Login	Landing Page → Sign In	Email + Password, then TOTP on next page
Reset Password	Login → Forgot Password	Requires TOTP
Edit Profile	Navbar → Profile	Click Edit on each section
Upload Resume	Navbar → Resume	PDF or DOCX format
Download Resume	Navbar → Resume	Requires TOTP
Search Jobs	Navbar → Jobs	Use filters for better results

Apply for Job	Jobs → Apply Now	Resume optional
Track Applications	Navbar → Applications	View status updates
Create Company	Navbar → Companies → Create	Recruiter only; page visible to all
Post Job	Companies → My Company → Post	Recruiter only
View Applicants	Companies → Job → Applicants	Recruiter only
Send Message	Navbar → Messages	E2EE encrypted
Find People	Navbar → Connections	Send connection requests
Export Keys	Profile → Encryption Keys	Save `.vkkey` file
Import Keys	Profile → Encryption Keys	Needs password + file
Change Password	Navbar → Security	Requires current PW + TOTP
Delete Account	Navbar → Security	Irreversible, requires TOTP
Admin Panel	Navbar → Admin Panel	Admin or Platform Admin role only
Report User	Messages → DM → Flag Icon	Select category + describe

Keyboard Shortcuts

Shortcut	Action
`Enter`	Send message (in Messages page)
`Enter`	Execute search (in Jobs and Connections pages)

Glossary

Term	Definition
TOTP	Time-based One-Time Password — a 6-digit code generated by your authenticator app that refreshes every 30 seconds; the server accepts each code for approximately 90 seconds (current ± 1 time-step)
E2EE	End-to-End Encryption — messages are encrypted on your device and only decryptable by the intended recipient
RSA-4096	A 4096-bit asymmetric encryption algorithm used for key wrapping (RSA-OAEP), message signing (RSA-PSS), and resume digital signatures (PKCS1v15)
AES-256-GCM	A symmetric authenticated encryption algorithm used for encrypting message content

PKI	Public Key Infrastructure — a framework for managing digital certificates and encryption keys
.vkkey	VajraKavach Key file — a JSON file containing your RSA key pair for E2EE backup/transfer
Virtual Keyboard	An on-screen keyboard with randomized digit positions to prevent keylogger attacks
Fernet	A symmetric encryption scheme (AES-128-CBC + HMAC-SHA256) used for encrypting resumes at rest
Hash Chain	A sequence of log entries where each entry's hash includes the previous entry's hash, creating tamper-evidence
Argon2id	A memory-hard password hashing algorithm resistant to GPU and ASIC-based cracking